

Volcanism on Mars

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Types,
Extent and
data how much
surface is volcanic

More than 15 years of planetary exploration of Mars have given insight into the geologic processes that have shaped its surface. The newly acquired Viking data have shown that volcanism is one of the most important geologic processes operating on Mars throughout its history. In situ chemical analyses of Martian soil by the Viking lander spacecraft indicate mafic to ultramafic source rocks. This is consistent both with available remote sensing data, which indicate the presence of mafic minerals such as pyroxene and olivine, and with petrologic modeling, based on available geophysical data which suggest that Martian lavas are probably iron rich and ultramafic. These data strongly suggest that basaltic volcanism is widespread on Mars, and much of the photogeological data may be studied in this context. Photogeological analysis of the Martian surface has shown two main types of volcanic morphologies: the first type is central volcanoes, which are volcanic landforms developed by continued and prolonged eruption from a point source vent. This category includes (1) shields, the classic low-profile volcanic mountains of which Olympus Mons is the most spectacular example, (2) domes, steep-sided constructs, such as Tharsis Tholus, that may represent lower rates of eruption than the shields or, possibly, more silicic lava compositions, (3) highland patera, radially textured low-profile volcanoes that occur in the cratered terrain and are interpreted as ash shields, (4) Alba Patera, an apparently unique volcanic landform consisting of a vast volcanic center over 1500 km across with flank slopes of less than a tenth of a degree, and (5) various small features such as cinder cones. The second major category is volcanic plains, which are units recognized by several criteria, of which the presence of mare ridges and flow lobes are the most useful. Volcanic plains are subdivided into four main groups: (1) simple flows, broad, smooth to rolling plains that contain numerous mare-type ridges but no flow lobes, interpreted as being composed of thick, single-cooling units, (2) complex flows, displaying multiple overlapping flow lobes interpreted to be indicative of thin, multiple-cooling units, (3) undifferentiated flows, plains that typically lack any morphologic identifying feature but are considered to be volcanic partly on the basis of their association with large volcanic centers, and (4) questionable plains, volcanic(?) units heavily modified by other processes (erosion, tectonism, etc.) so that their origins are uncertain. When these categories of volcanic morphologies are combined with relative age data provided by crater statistics, a volcanic history for Mars can be derived as follows: Early heavy bombardment of Mars was accompanied and followed by small-scale fluvial channeling, extensive flood volcanism (the plateau plains), and ash shield volcanism in the cratered terrain. Shortly after this time, less extensive flood volcanism continued to resurface the planet during formation of the northern/southern hemisphere dichotomy. Central volcanism became more prominent with the development of the Alba Patera center as well as the older shields and domes of the northern hemisphere (early Tharsis and Elysium regions). The development of the Tharsis and Elysium uplifts may have triggered the release of large-scale catastrophic floods, producing large channels. Continued uplift and lithospheric thinning concentrated volcanic activity in the Tharsis region, producing large shield volcanoes and extensive lava plains. Both central vent and plains volcanism have been active throughout Martian history, but the volumes of extrusion have gradually decreased with time. This is consistent with a moonlike thermal history involving a lithosphere of increasing thickness with time, gradually 'turning off' the volcanism. Although many questions remain regarding Martian volcanism, the Viking data have provided a remarkable, detailed overview of the probable nature of the volcanic history of Mars.

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A. INTRODUCTION

Throughout the past decade and a half of lunar and planetary exploration, volcanism has been found to be an extremely important process in the evolution of the terrestrial

planets. Mariner 9 first returned conclusive evidence for volcanism on Mars (Figure 1) with spectacular images of shield volcanoes and lava flows [McCauley *et al.*, 1972]. Detailed study of Mariner 9 images showed that substantial parts of the planet are covered by volcanic material, although the definite identification of some of the units remained questionable. During the post-Mariner-9, pre-Viking period, numerous topical studies of Martian volcanoes were carried out, principally on the shield volcanoes [e.g., Carr, 1973, 1975; Blasius, 1976; Malin, 1977] but also including analyses of volcanic channels [Carr, 1974; Greeley, 1973] and possible cinder cones [West, 1974]. A synthesis of the volcanic mapping [Spudis and Greeley, 1977; Scott and Carr, 1978] showed that as much as 54% of the planet is covered with probable volcanic materials.

In assessing volcanic activity on Mars the following questions should be addressed: (1) What is the geographic distribution of volcanic features and units? (2) What are the different types of volcanoes and what styles of volcanism were involved in their formation? (3) When and over what length of time were the various volcanoes formed? (4) Where are the vents

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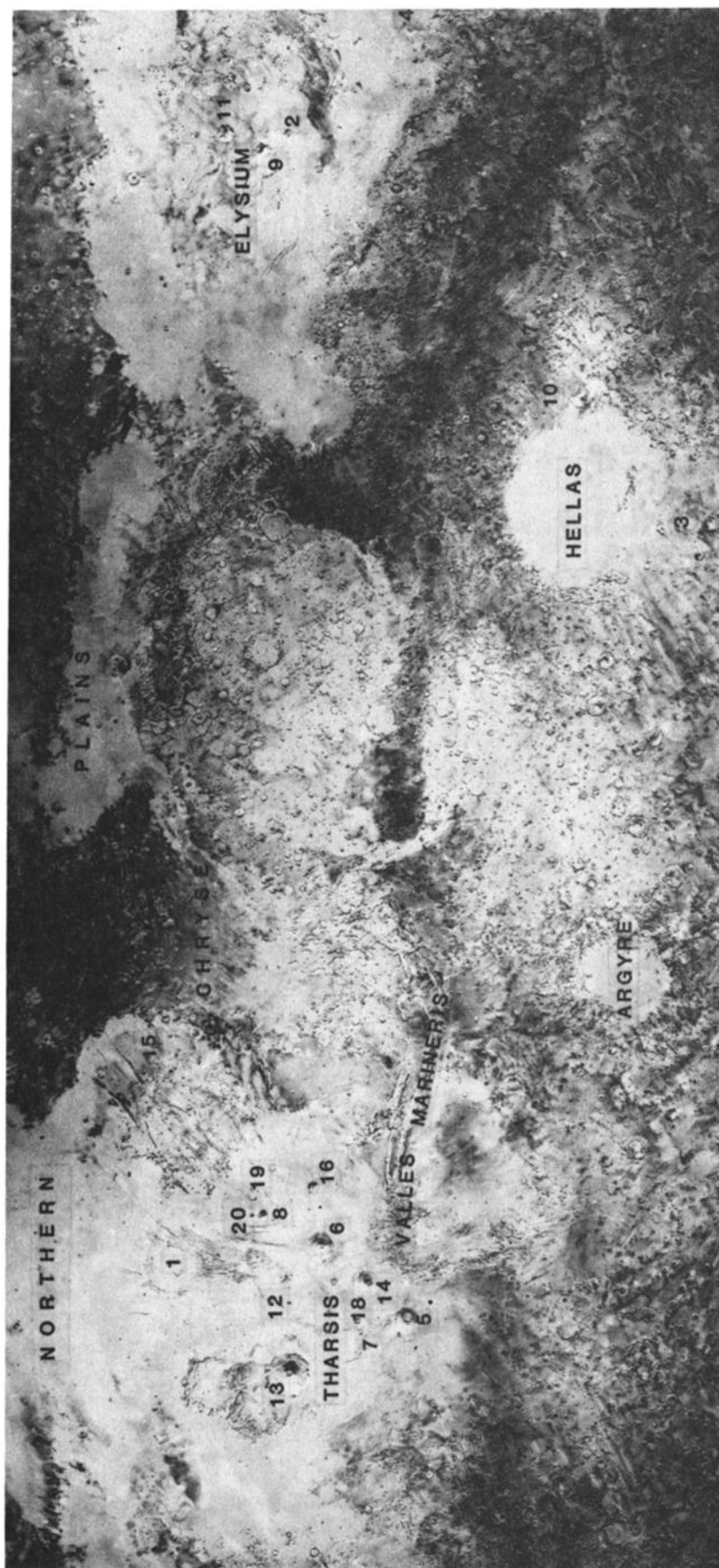


Fig. 1. Shaded airbrush relief map of Mars to 65°N and 65°S latitude, showing named volcanoes and other prominent features: 1, Alba Patera; 2, Albor Tholus; 3, Amphitrites Patera; 4, Apollinaris Patera; 5, Arsia Mons; 6, Ascraeus Mons; 7, Biblis Patera; 8, Ceraunius Tholus; 9, Elysium Mons; 10, Hadraica Patera; 11, Hecates Tholus; 12, Jovis Tholus; 13, Olympus Mons; 14, Pavonis Mons; 15, Tempe Patera; 16, Tharsis Tholus; 17, Tyrrhena Patera; 18, Ulysses Patera; 19, Uranus Patera; and 20, Uranus Tholus. (Base map courtesy of U.S. Geological Survey.)

for the volcanic materials? (5) What are the chemical and physical properties of the volcanic materials? (6) What was the origin of the magma bodies involved in the volcanism? (7) How do other geological events relate to the episode(s) of volcanism? The Viking mission provides a global high-resolution photographic data base of Mars and allows a means of addressing some of these questions.

In this paper we review the current state of knowledge of volcanism on Mars and attempt to synthesize the available geochemical, geophysical, and photogeological data into a general picture of Martian volcanism. Emphasis will be placed on photogeological data. We present a classification of Martian volcanic features and show their global and temporal distribution (Figures 1–3). From comparisons with terrestrial and lunar analogs we interpret the styles of volcanism represented on Mars. The classification, mapping, and interpretations of volcanic style are then combined with a Martian time scale to derive a generalized volcanic history for Mars. From this history we hope to provide a framework for comparing volcanic episodes with other events in the geological history of Mars. Finally, we identify the questions left unanswered in regard to Martian volcanism.

1. Data Base

The study of Mars has reached a turning point from essentially a period of reconnaissance, data accumulation, and relatively rapid interpretation to a period of methodical data analysis and synthesis. Exploration of Mars has gone through the flyby (Mariners 4, 6, and 7), orbiter (Mariner 9, Viking Orbiters 1 and 2), and the lander stages (Viking Landers 1 and 2). With the termination of the orbiter phase of the Viking mission in 1980 after more than 4½ years of highly successful operation, possible additional global data will not be available until the flyby of the Galileo spacecraft in the mid-1980's. Thus with the exception of earth-based observations, we now have all the data for Mars that we are likely to acquire for at least the next 5 years.

As relevant to studies of volcanism, the photogeological data base consists of more than 8000 moderate- to low-resolution (~0.1 to 3 km) images obtained by Mariner 4, 6, 7 and 9, more than 50,000 images returned by the Viking orbiters (some with resolution better than 10 m), and images taken on the surface by the Viking landers. With the exception of the northern plains and some areas of the southern cratered terrain, contiguous moderate-resolution (~25–50 m) images cover most of the planet and permit regional photogeological assessment of its surface. Unfortunately, because of lack of uniform coverage, the northern plains remain largely unknown not only for the volcanic history but in regard to the general geological history of Mars as well [Scott, 1979].

Data on the composition of Martian surface materials include three sources: Viking landers, Viking orbiters, and earth-based observations. Viking lander information comes primarily from the X ray fluorescence experiment and studies of the spectral and physical properties of the Martian soil. Global compositional data are derived from color filter images obtained by the Viking orbiters; although high in spatial resolution, the spectral resolution is rather coarse for compositional analyses. Earth-based spectral observations, on the other hand, are relatively high in spectral resolution but poor in spatial resolution; thus attempts have been made to com-

bine data sets from earth-based observations and the Viking orbiters to provide a better evaluation of regional surface materials.

The geophysical data base for Mars is extremely limited. Studies of the moment of inertia and other gravity data draw on analyses of spacecraft orbits coupled with rather coarse topographic data. Seismic data obtained by Viking Lander 2 [Anderson *et al.*, 1977] is regarded as poor primarily because the instrument is not well coupled with the ground. Estimates of interior characteristics, such as the properties of the Martian mantle, are derived through various theoretical models and remain open to question.

A substantial literature has grown for the general geology of Mars, including discussions of Martian volcanism. An excellent summary of geological knowledge prior to the Viking mission is by *Mutch et al.* [1976]. Viking results are found in special issues of the *Journal of Geophysical Research* (September 1977 and December 1979) and *Icarus* (June 1978); in addition, the general geology of Mars based on Viking data has been reviewed by *Carr* [1980] and *Arvidson et al.* [1980]; an atlas of possible earth analogs is given by *Carr and Greeley* [1980].

2. Approach

We first review the available data on the composition of surface materials on Mars and discuss the models of lavas and other rocks derived therefrom. We then review the implications that such rock types would have for the physical properties (e.g., viscosity) of the lavas and the styles of volcanism involved in their emplacement.

The primary data set for analyses of Martian volcanism is photogeological. Thus our main approach here is to classify the various volcanic features identified on Mars, map their distribution, and place them in a relative time sequence. The classification scheme used is based primarily on surface morphology; because the morphology of volcanic landforms is mainly a function of the style of volcanism, this scheme will enable interpretation of the volcanic processes involved in their formation. Relative dating of the mapped volcanic features is based on superposition, crosscutting relations, and impact crater statistics.

B. COMPOSITIONAL CONSIDERATIONS FOR MARTIAN VOLCANISM

Estimates of the composition of volcanic materials on Mars are derived from analyses of surface materials at the Viking landing sites, from remote sensing data obtained by the Viking orbiter and earth-based observations, and from theoretical modeling based primarily on geophysical and petrological considerations.

1. Viking Lander Data

Both Viking landers are sited in the northern hemisphere on plains units generally regarded from orbital photogeology as volcanic. Information on the composition of surface materials is derived from the X ray fluorescence spectrometer, experiments dealing with the physical properties of surface materials, and from color imaging data. Rocks at both sites show abundant holes and pits (Figure 4) regarded by some investigators as vesicles in volcanic rocks [e.g., *Binder et al.*, 1977], al-

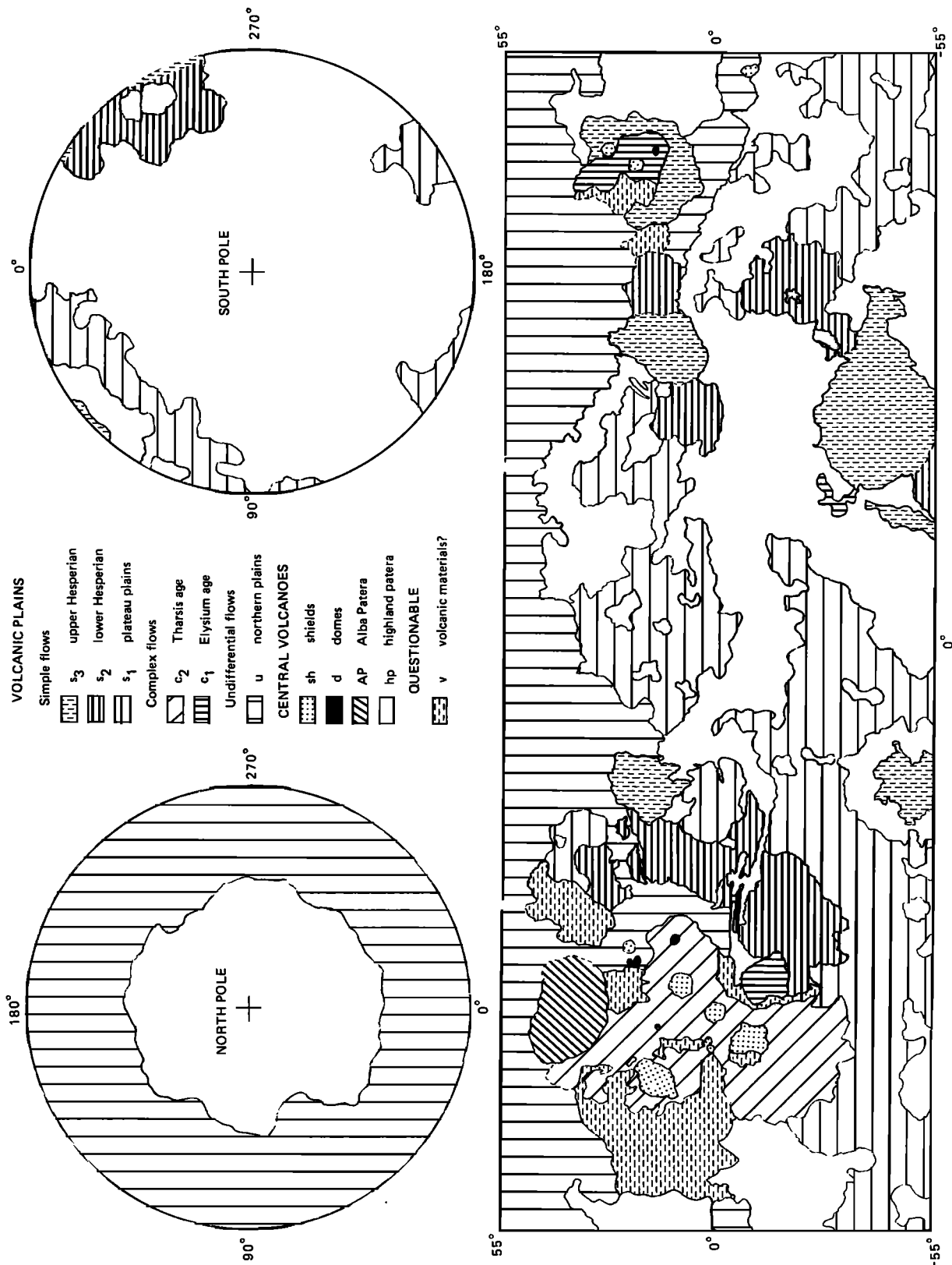


Fig. 2. Photogeological map of volcanic units on Mars by morphologic type. As is the case with volcanic regions on earth, more than one type of volcanic feature may occur in the mapped units on Mars; categories shown here are based on the prevailing type of feature for each region.

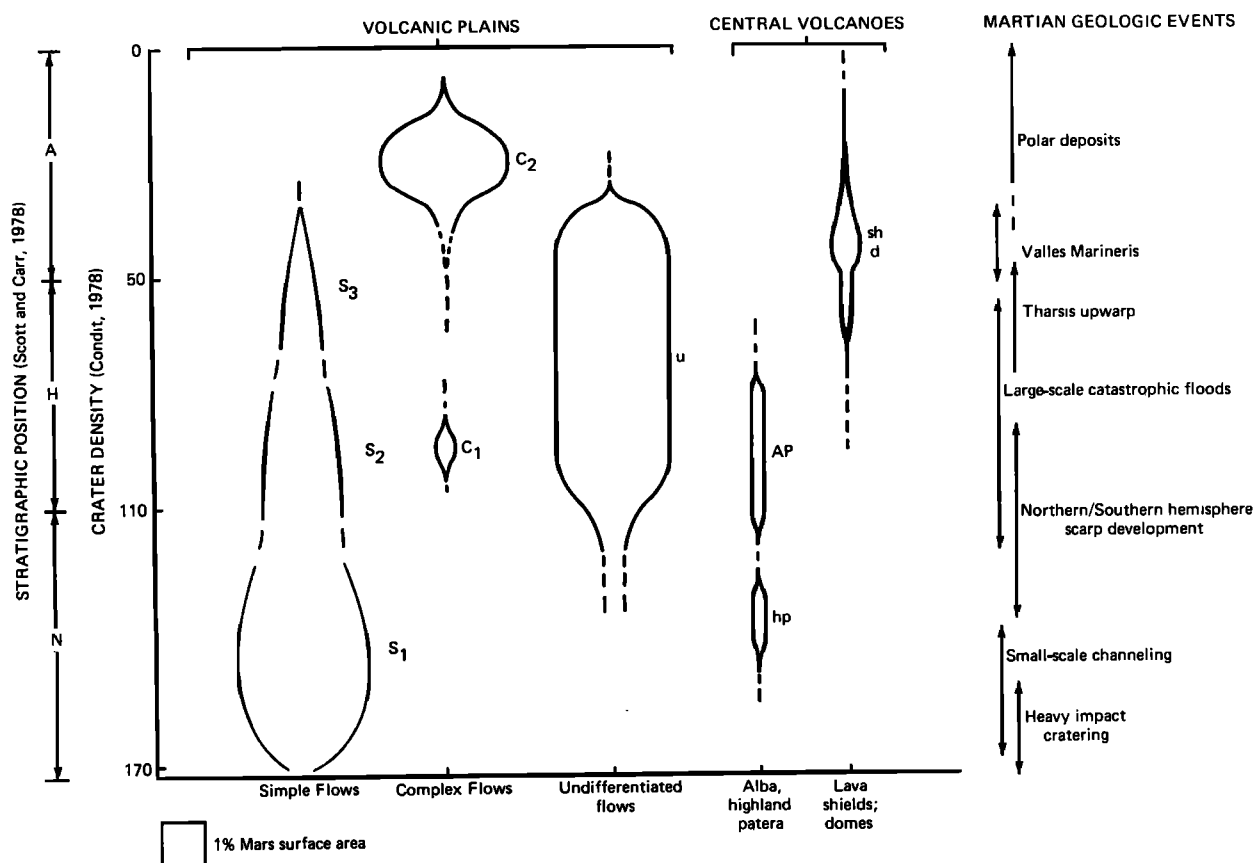


Fig. 3. A summary of the volcanic history of Mars derived from photogeologic study. The envelopes are proportional in area to the currently exposed areas on Mars of the respective volcanic units. Relative age data are derived from stratigraphic relations, comparison with stratigraphic data of *Scott and Carr* [1978], and correlation with the density of 4- to 10-km craters where good data are available [Condit, 1978]. There is no implication of absolute time in this diagram. The lengths of arrows representing Martian geologic events are related to uncertainty in the placement of the event and do not necessarily indicate its duration. This diagram is based on a similar concept derived to summarize lunar geologic history by D. E. Wilhelms in the work of *Masursky et al.* [1978].

though other investigators have suggested that similar features could be produced by nonvolcanic processes such as aeolian erosion [McCauley *et al.*, 1979].

More than 22 samples [Clark and Baird, 1979] at both landing sites have been collected and analyzed by X ray fluorescence techniques. In all cases the results are nearly identical with the exception of sulfur. The Martian samples are grossly similar in composition to lunar and terrestrial basalts in that they appear to be derived from mafic source rocks, but differences between the planets are significant (Table 1). Despite efforts to collect a variety of materials for analyses by the Viking landers it is generally regarded that no crystalline rocks have been sampled but rather that the samples consist of relatively homogeneous lumps of partly consolidated, weathered soils, an interpretation enhanced by their low bulk densities (1.2 g cm^{-3} [Clark *et al.*, 1976]). Analyses of the magnetic and physical properties of the soils show a high proportion of magnetic mineral grains (probably magnetite or maghemite) among the 'fines' at both sites [Hargraves *et al.*, 1977, 1979], which would partly account for the high iron content of the soils. Clark [1979] summarizes four interpretations of the compositions of the samples:

1. The materials consist primarily of nontronite (an iron-rich montmorillonite clay); on earth, nontronite is a product of weathering of mafic igneous rocks; although rare in the weathering of basalts on land, they are a major product of

subaqueous hydrothermal alteration in seafloor spreading centers.

2. The materials consist of a mixture of magnetic minerals, powdered picrite basalt, and leached salts.

3. The materials are derived from ferropicritic ultramafic lavas from the Martian mantle.

4. The materials consist of a mixture of finely divided basaltic powder, primitive materials of planetary accretion, and condensates of volatiles released by volcanism.

Regardless of interpretation, all models involve mafic to ultramafic parent materials, probably in the form of basaltic rocks. Although spectral data obtained by the lander cameras show a variety of colors in the fine surface materials [Huck *et al.*, 1977; Jobson *et al.*, 1978], and the X ray fluorescence results indicate subtle changes in composition, these variations do not alter the interpretation of the parent materials being mafic to ultramafic volcanic rocks.

2. Remote Sensing Data

Earth-based telescopic observations of Mars to map spectral regions have been carried out for more than a decade, summarized by Singer *et al.* [1979]. These observations, principally in the visible and near-infrared (0.3 to $2.6 \mu\text{m}$), have been combined with multispectral images obtained from the Viking orbiters [Soderblom *et al.*, 1978] to determine surface variability on as nearly a global scale as is currently possible.



Fig. 4. Viking Lander 2 image showing rocks that contain numerous voids; voids have been interpreted as vesicles indicative of volcanic origin and, alternatively, as pits resulting from aeolian abrasion; largest rocks in view are about 40 cm across (NASA-IPL PIC ID 77/03/23/031154).

Although all spectra are characterized by strong Fe^{3+} absorptions, dark areas (low albedo) show this effect less and are interpreted to consist of less oxidized materials. Dark areas also show the greatest internal variability and have Fe^{2+} absorption near $1.0\ \mu\text{m}$, attributed primarily to the presence of pyroxenes with less than 5% olivine [Singer, 1980]. Thus dark areas are considered to be somewhat oxidized basaltic or ultramafic rock of regionally variable composition. Bright areas are considered to be fine grained assemblages of mineral hydrates, ferric oxides, and minor amounts of relatively unaltered mafic materials; this would be essentially the chemical and physical weathering products of basaltic or ultramafic parent materials (e.g., materials sampled at the lander sites).

Thus on a global scale the dark areas of Mars appear to consist of mafic to ultramafic igneous rocks whose weathering products contribute to the formation of bright regions.

3. Compositions Derived From Geophysical Modeling

An estimate of the composition of Martian lavas was made by McGetchin and Smyth [1978] based on predicted mineral assemblages for the Martian mantle. Beginning with an estimate for the density of the Martian mantle given by Reasenberg [1977] as $3.55\ \text{g cm}^{-3}$, and with a 'starting' composition of pyrolite (based on terrestrial mantle composition), McGetchin and Smyth used a modified CIPW normative scheme and added FeO to produce the appropriate density for Mars. The resulting assemblage is not garnet lherzolite as in the earth but is an oxide-garnet wehrlite with no orthopyroxene. Partial melting of this material would produce iron-rich, ultramafic lavas of very low viscosities described by McGetchin and Smyth as picritic alkali basalts that would have erupted at high rates of effusion and been emplaced as flood basalts. This composition is nearly the same as that proposed by Made-

razzo and Huguenin [1977] based on a model derived from the Viking lander X ray fluorescence results, reflection spectroscopy, and a photochemical weathering model [Huguenin, 1974]. However, a recent revision in the estimate of the Martian mantle density ($3.44\ \text{g cm}^{-3}$ [Goettel, 1980] would probably tend to yield a more terrestriallike mantle mineral assemblage (Fe-rich lherzolite) producing less exotic basaltic magmas during partial melting. This question will probably remain open to debate until the acquisition of higher-quality remote sensing data on Martian lava compositions.

In summary, several independent lines of evidence point toward igneous rock compositions that are mafic to ultramafic in character. Global spectral mapping suggests that dark areas may represent exposures relatively free of fine grained aeolian sediment and that there may be considerable variation in composition within the generally iron-rich classes. Bright regions (and bright materials at the lander sites) appear to consist, in part, of weathered, windblown clay minerals derived from the weathering of mafic parent materials; some models allow the clays to be derived from phreatomagmatic events.

C. PHYSICS OF MARTIAN LAVAS

Several studies in recent years involve attempts to relate lava flow morphology to both rock chemistry and eruption mechanism through considerations of the rheological properties of the lavas. This technique has great potential in that valuable geological and geochemical information may be gathered primarily through photogeologic interpretation. This assumes, however, that the physical and chemical parameters controlling lava flow morphology are well understood.

One of the first attempts to relate lava morphology to eruption conditions in a quantitative manner was the study of terrestrial lava flows by Walker [1973], who found that although

TABLE 1. Chemical Composition of Martian, Lunar, and Terrestrial Samples

Oxide, wt. %	Sample*					
	1	2	3	4	5	6
SiO ₂	44.7	42.8	44.48	45.03	37.79	49.34
Al ₂ O ₃	5.7	n.a.	11.25	7.27	8.85	17.04
FeO	...	...	11.38	21.09	19.66	6.82
Fe ₂ O ₃	18.2	20.3	3.00	...	...	1.99
MgO	8.3	n.a.	17.32	16.45	8.44	7.19
CaO	5.6	5.0	9.54	8.01	10.74	11.72
K ₂ O	<0.3	<0.3	0.40	0.06	0.05	0.16
TiO ₂	0.9	1.0	n.a.	2.54	12.97	1.49
SO ₃	7.7	6.5	...	...	...	...
Cl	0.7	0.6	...	...	...	...
Sum	91.8	n.a.	98.37	100.45	98.50	95.75

Initials n.a. mean not available.

*Samples are the following: 1, Martian sample S1, Chryse Planitia [Toulmin *et al.*, 1977]; 2, Martian sample U1, Utopia Planitia [Toulmin *et al.*, 1977]; 3, model Martian lava, calculated composition [McGetchin and Smyth, 1978]; 4, lunar mare basalt, Apollo 12 olivine normative 12009 [Papike *et al.*, 1976]; 5, lunar mare basalt, Apollo 17 high-Ti 70215 [Papike *et al.*, 1976]; and 6, terrestrial basalt, oceanic tholeiite [Engel *et al.*, 1965].

gross chemistry of lava has some effect on flow dimensions, it is primarily the rate of effusion that controls the lengths of flows, with high eruption rates producing longer flows. These results were incorporated with general rheologic theory to the study of the young, well-preserved Mare Imbrium flows on the moon [Schaber, 1973; Moore and Schaber, 1975]. The lunar mare flows are known to be basaltic in chemistry, and some of their rheologic properties are well defined by studies of returned samples [e.g., Murase and McBirney, 1970]. The Mare Imbrium flows are extremely long (up to 1200 km) and flowed down a very gentle topographic gradient ($\sim 0.1^\circ$). Moore and Schaber [1975] conclude that lunar lavas behave as Bingham bodies in which the yield strength is an important parameter, and this assumption has been maintained in the studies of lava flows on Mars.

The earliest rheologic study of Martian lavas was the analysis by Hulme [1976] using Mariner 9 data for the Olympus Mons flows that have leveed channels. Although data are poor and the estimates of yield strength for these flows are crude, it was concluded that the Olympus Mons flows have flow properties comparable to lavas slightly more silicic than Hawaiian basalts, possibly as high in silica as terrestrial andesites. More recent studies of Martian lavas using Viking data [Schaber *et al.*, 1978] suggest lavas very similar in rheologic properties to the Mare Imbrium flows on the moon, i.e., basaltic lava flows. This is more consistent with spectral data and geophysical modeling [e.g., McGetchin and Smyth, 1978] suggesting the predominance of basaltic volcanism on Mars. One possible problem with this interpretation is discussed by Moore *et al.* [1978], challenging the hypothesis that lava flows behave as Bingham bodies. Because the effects of such variables as pre-flow surface roughness and percentage of entrained solids are poorly known at best, it seems wise not to take lava flow morphologic data too far in speculating about the details of volcanic chemistry. It seems safe to say, however, that Martian lava flows display rheologic behavior that is consistent with general basaltic chemistry.

The viscosity of Martian lavas may be estimated by several indirect means. The petrologic modeling of McGetchin and Smyth [1978] suggested that partial melts of the postulated Martian mantle would produce ultramafic lavas of extremely low viscosity, less than 10 P for Martian lavas, a factor of 50 to 100 below terrestrial Hawaiian basalts. A recently revised esti-

mate of Martian mantle density [Goettel, 1980] and refined spectroscopic data for Martian surface materials [Singer, 1980] suggest that Martian lavas are not quite as mafic as suggested by the McGetchin and Smyth [1978] model (approximately 5% versus 40% modal olivine in Martian lavas). However, the data are consistent with lava chemistries grossly similar in bulk composition to lunar mare basalts. Because the rheologic properties of Martian lava flows appear similar to lunar mare lava flows, it may be inferred that the lavas probably have similar viscosities at extrusion, of the order of 5–10 P at liquidus temperatures [Murase and McBirney, 1970]. These values are in agreement with those derived by Schonfeld [1979] as 3, 5, and 60 P for three flows in Arsia Mons based on estimated rates of effusion—derived from flow lengths—and yield strengths.

The lengths of Martian lava flows are quite variable, with some individual flows being more than 300 km long. Such long lengths strongly suggest extremely high eruption rates [Walker, 1973] as well as low-viscosity magmas (see section D2b). On the moon, such eruption conditions typically produce featureless, flat mare flood basalts. On Mars, both flood lavas and central vent volcanism have occurred throughout its history, and the presence of long flows lends support to the idea of high eruption rates.

In summary, the physical properties of Martian lavas as derived primarily from photographs suggest extensive basaltic volcanism of a type similar to that seen on the moon. In addition, some of the unusual Martian central vent flows suggest unique styles of volcanic eruption not seen in either lunar or terrestrial volcanic complexes. The variety of evidence from photogeologic, geophysical modeling and experimental petrology data support the idea that Martian lavas were very fluid, comparable to the lunar mare basalts. Detailed analysis using new Viking orbiter topographic data currently being derived [Wu, 1979] will improve our understanding of the physics of Martian lavas and hence our knowledge of their chemistries and probable eruption conditions.

D. PHOTOGEOLOGICAL ANALYSIS

1. General Statement

Nearly all knowledge of the geology of Mars has been derived from photogeological studies. Near-global photographic coverage at moderate to low resolution (~ 0.1 to 3 km) was ob-

TABLE 2. Classification and Extent of Martian Volcanic Features

Type	Symbol	Example	Characteristics	Extent 10 ⁶ km ²	Percent Mars Surface
<i>Central Volcanoes</i>					
Alba	AP	Alba Patera	unique, extremely low relief shield-like volcano; sheet and tube-fed flows	1.13	0.78
Highland patera	hp	Tyrrena Patera	low relief, degraded, radially textured volcanoes	0.23	0.16
Shields	sh	Olympus Mons	broad, moderate relief central vent volcanoes; tube-fed flows	1.23	0.85
Domes	d	Tharsis Tholus	steeper-sided central volcanoes	0.08	0.05
Total central volcanoes				2.67	1.84
<i>Volcanic Plains</i>					
Simple flows	s	Hesperia Planum	regional plains; wrinkle ridges only; no flow lobes	42.08	29.22
Complex flows	c	Tharsis Plains	complex flow units; flow lobes abundant; rare wrinkle ridges	9.16	6.36
Undifferentiated	u	northern plains	plains of uncertain origin (flow lobes rare), although probably volcanic	27.37	19.00
Questionable	...	aureole materials	modified units associated with volcanic units	5.92	4.11
Total volcanic plains				84.53	58.69
Total volcanic surface area on Mars				85.20	60.53

tained by Mariner 9 in 1971–1972 which provided the data base for global studies, including the Mars Geological Mapping Program. This program resulted in the publication of 30 1:5,000,000 scale geological maps by the U.S. Geological Survey and a smaller-scale (1:25,000,000) synoptic geological map for the whole planet [Scott and Carr, 1978]. Most of the general approach and techniques used in planetary geology have come from the study of the moon. The moon has served as a good training ground because it is relatively simple as compared to the larger terrestrial planets. In addition, we have had the benefit of ‘field checking’ some of the photo-geological interpretations through the Apollo missions. Although Mars, partly because of its greater size and the presence of an atmosphere, is more complex than the moon, the same general methods of study can be applied.

The Viking orbiter images provide a significant increase in detail over previous photographs of the surface and allow a much better assessment of contact relationships between geological units and of the geological processes that have shaped the surface. Consequently, not only is the general geology being remapped in light of Viking data, but the major and minor geological processes that have operated on Mars are being reevaluated, including reassessment of the effects of volcanic processes and history. Volcanism has been dominant throughout much of the history of Mars; therefore volcanic features and interpretations of volcanic processes are also being reassessed through Viking data. Because the primary data base is photogeological, our approach must be based on the morphology of suspected volcanic features.

The morphology of volcanoes [e.g., Macdonald, 1972; Cotter, 1969] is the result of many complex and often interrelated parameters, which collectively define the style of volcanism. These parameters include the physical and chemical properties of the erupting magma and related products, the size, shape, and arrangement of the vent(s), the environment into which materials are erupted, and the topographic and structural setting of the volcanic activity. Thus the morphology

provides clues to these factors and photogeological studies can provide insight into the styles of volcanism. This approach in interpreting volcanic history, however, is far from definitive. Most terrestrial and lunar volcanic interpretations based on morphology have dealt primarily with basaltic features; first, because basaltic volcanism is predominant on the moon, and second, the morphology of pristine silicic volcanic features is very poorly known even on earth. Most well-preserved silicic features on earth are of limited areal extent. Larger silicic features such as extensive ash sheets are geologically older features that have been degraded and do not display primary surface morphologies. Thus although silicic volcanic features may occur on Mars, at present we do not know their identifying characteristics. A similar problem occurs in regard to basaltic ash deposits. Because basaltic ash flows are relatively rare on earth, we are ignorant of their surface morphology; yet, such deposits may be important on other planets, particularly on Mars, as will be discussed below.

Volcanoes on earth typically are classified by their morphology (shield shaped, dome, etc.), the size, shape, and arrangement of their vents (e.g., central vent versus fissure vent), or on their style of origin (e.g., ‘monogenetic’ versus ‘polygenetic’). Because it is difficult and often impossible to determine vent characteristics and mode(s) of origin, classification of volcanoes on the moon and planets is based on general morphology. On Mars, this classification is essentially two-fold: central volcanoes and volcanic plains. As these terms imply, central volcanoes include features that involved essentially ‘point source’ vents around which volcanic materials accumulated to form shields, domes, and other constructs. Volcanic plains include widespread, relatively flat lying units, with or without flow fronts and with other features indicative of volcanic origin. Source vents for the plains are often extremely difficult or impossible to determine and are often assumed to be of the fissure (linear vent) variety on the basis of analogy with terrestrial lava flows. Table 2 gives the classification scheme employed for volcanic features on Mars.

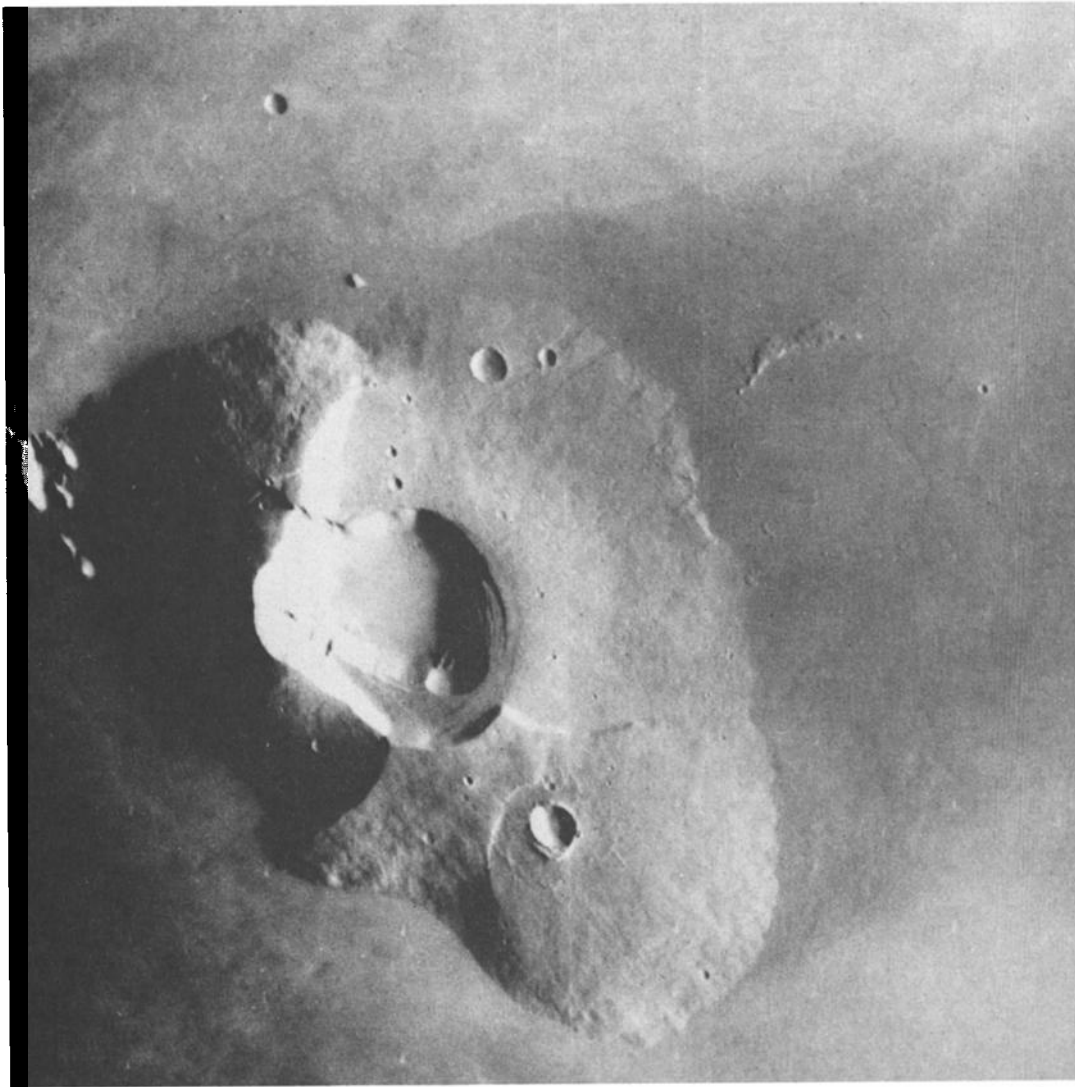


Fig. 5. Tharsis Tholus, a 110×170 km central vent volcano, has steep flanks and is classified as a dome. Younger lava flows erupted from elsewhere in the region and partly bury the dome; the dark zone east (to the right) of the dome appears to be a wind streak formed in the lee of the volcano. North is to the top; illumination from the right (Viking orbiter image 858A23).

2. Central Volcanoes

Mariner 9 images show a wide variety of volcanic and possible volcanic features of the central type which were initially described primarily on the basis of their topographic profiles and include (1) tholii—domes with steep slopes (Figure 5), (2) shields—the classic central volcanoes such as Olympus Mons (Figure 6) similar in form to shield volcanoes on earth, (3) patera—a collective term for a variety of unusual ‘saucer-shaped’ features displaying channels and radial surface markings and which often have central calderas (Figures 7 and 8), and (4) a host of small (~ 1 km) positive relief features thought to be cinder cones (Figure 9) or other small volcanoes (Figure 10). Attempts were made to relate the morphology of these various Martian features to similar appearing volcanoes on earth; thus the Martian shields were compared to shields on earth and were interpreted to be composed essentially of basaltic lavas; tholii, being steep domelike features, were compared to volcanic domes on earth and were considered to represent more viscous lavas, possibly of silicic compositions; the patera have no terrestrial (or lunar) counterparts.

It was considered, on the basis of analysis of Mariner 9 data, that the height of central vent volcanoes was inversely proportional to geologic age [Carr, 1976] based on a model in which the lithosphere increased in thickness with time, generating higher hydrostatic pressures to force lava to greater elevations (e.g., the Olympus Mons summit at 27-km elevation). It appears from Viking data, however, that some Martian volcanoes do not easily fit into this scheme, such as Alba Patera, an extremely low relief, intermediate age structure that precludes a simple relationship between volcano height and relative age.

Viking orbiter images now show far superior detail for most of the volcanic features than was first seen through Mariner 9. Not only is it now possible to analyze the individual flows composing many of these features, but the stratigraphic relations of the volcanoes to the surrounding terrain can also be determined. For example, many of the features named patera and tholii from Mariner 9 images are now seen to be simply modified or partly buried shield volcanoes (Table 3 and Figure 11). This poses a dilemma in which Martian nomenclature and geological classification are unfortunately tangled. In the

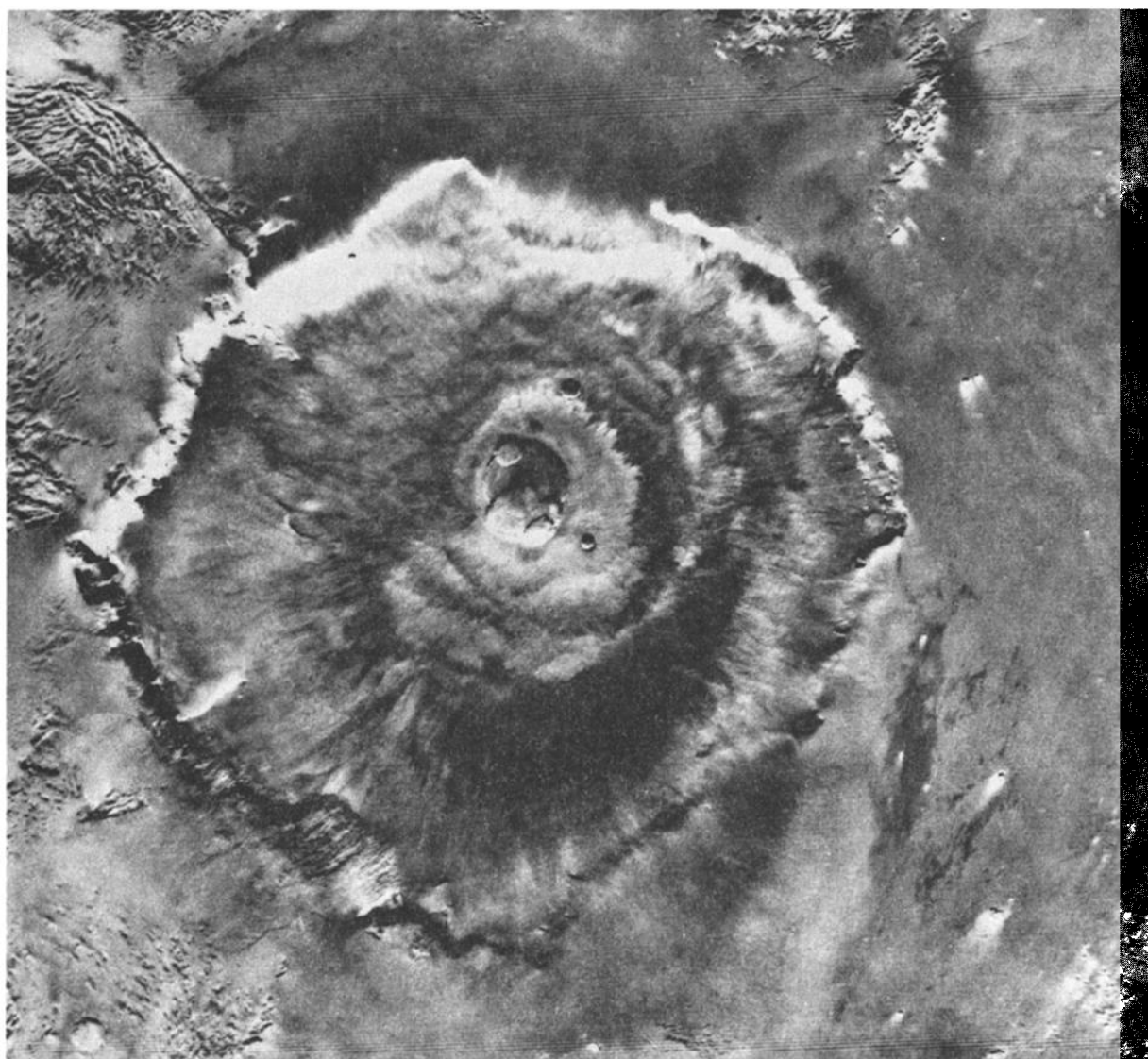


Fig. 6. Olympus Mons, one of the best known shield volcanoes in the solar system, is more than 600 km across and is nearly completely encircled by a prominent scarp. High-resolution Viking orbiter images show that flows erupted from Olympus Mons spilled over the scarp and flowed hundreds of kilometers onto the surrounding plains. North is to the upper left; illumination from the top (Viking orbiter image 649A28).

strict sense, the terms patera and tholii are applied to features having a certain form in the same sense that 'mountains' and 'valleys' are terms applied to landforms without implications of geological classification. Thus many Martian features formally named as patera and tholii can have markedly different origins. For example, Orcus Patera appears to be a highly modified impact crater, probably not involving volcanic processes, while Biblis Patera appears to be an older shield volcano in the Tharsis region that has been nearly buried by younger plains-forming lava flows.

In the following classification scheme the subcategories of central constructs begin with low-profile features and progress through categories having steeper flank slopes.

a. Alba Patera. Alba Patera (Figure 12) represents an apparently unique volcanic landform, because it has no known counterpart on earth, moon, or Mercury [Carr *et al.*, 1977; Greeley *et al.*, 1978]. Thus it constitutes a separate class within the central vent volcanoes. Even on the relatively poor Mariner 9 images, Alba Patera was seen to be more than 1000 km across, making it the largest central vent volcano of the

planets explored thus far. Alba Patera is characterized by a set of ring fractures which encircle the central caldera and are part of a regional, Tharsis-related, NE-SW set of fractures. Originally called the Arcadia Ring [Carr, 1973], the set of fractures are formally named Alba Fossae on the west side and Tantalus Fossae on the east side. The ring fractures have a diameter of about 600 km, nearly equal to the diameter of Olympus Mons. Mariner 9 images show a suggestion of radial flows, and because most of the fractures were seen to cut the flows, Alba Patera was interpreted to be a rather old volcanic feature that had been degraded and tectonically modified. Carr [1973] suggested that it might have been an enormous shield volcano that had essentially collapsed to its present low profile. Evidence of degradation was provided by a single high-resolution Mariner 9 frame which showed numerous channellike features arrayed in a dendritic pattern, interpreted by many investigators to be fluvial in origin. Viking images show that these channels are not fluvial but are lava channels that reflect preserved flow morphology. Although Viking images confirm earlier interpretations that most of the

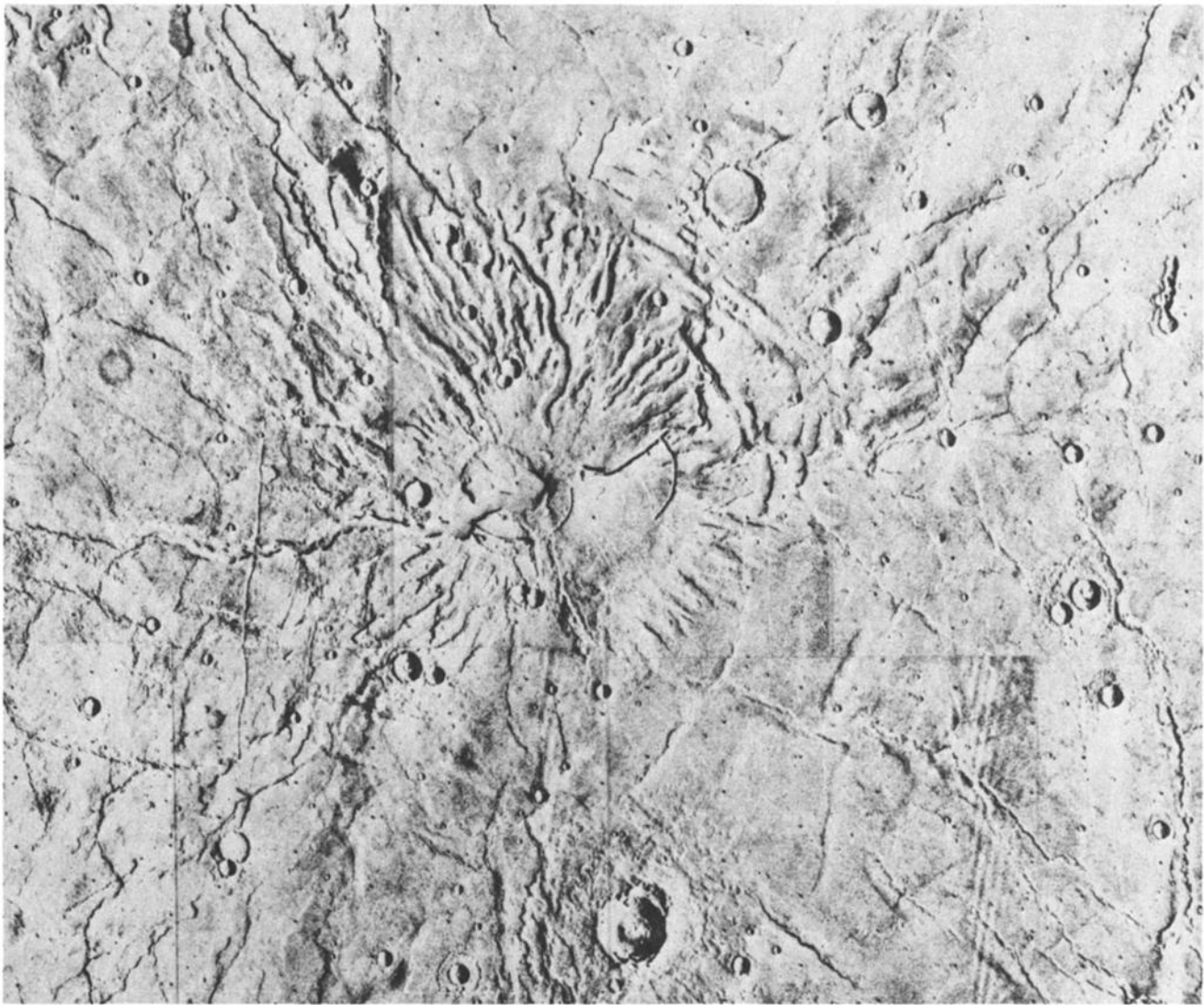


Fig. 7. Mosaic of Viking Orbiter 1 images of Tyrrhena Patera, one of the highland patera associated with the Hellas Basin in the southern hemisphere (Viking Orbiter 1 frames 87A12-17).

Alba structure was emplaced prior to tectonic deformation, some flows are also seen to be superposed over the fractures, indicating that volcanic activity continued to a minor degree during and after regional tectonic deformation [Wise *et al.*, 1979a].

One of the most striking aspects of Alba Patera is the diversity and relative crispness of the lava flows that make up the structure. Four main types of lava flows [Carr *et al.*, 1977] are identified: (1) tube-fed flows, (2) tube-channel flows, (3) sheet flows, and (4) undifferentiated flows. Tube-fed flows are characterized by ridges that tend to be radial to the center of Alba Patera. Lava channels and partly collapsed lava tubes (Figure 13) can be traced down the axes of the flows that were the main conduits feeding the advancing flow front. From terrestrial and lunar experience, the only commonly erupted lavas that have rheological properties conducive for lava tubes to develop are basalts. More silicic lavas are too viscous, and from observations of some active basalt flows, extremely fluid lavas also seem not to develop tubes. Rather, lava tubes develop best in basaltic lavas that are erupted over a long dura-

tion but at lower rates of effusion than those producing flood eruptions [Greeley, 1977]. Thus most terrestrial lava tubes develop from point source eruptions of basaltic lavas. Application of these relationships to Alba flows is consistent with both the central vent character of the volcano and consideration of the gross composition of its lavas. However, the Alba flows extend over very long distances, the longest tube-fed flow identified being more than 340 km. The average width for this and other tube-fed flows is about 8 km and is remarkably uniform. This width, however, represents only the exposed part of the flow. From experience with terrestrial volcanoes, lava channels and tubes frequently form constructional arches along their axes, principally by overflow, that act as topographic barriers [Greeley, 1971]. Subsequent flows bury the lower flanks of the arch but may not be able to breach the arch; the same relationships appear on Alba Patera (Figure 13), and therefore it is not possible to determine the actual width (exposed plus buried parts) of the tube-fed flows. However, taking the exposed part of the flows as a minimum and estimating the height from grazing sun incidence angles, a

GEOLOGIC MAP OF TYRRHENA PATERA

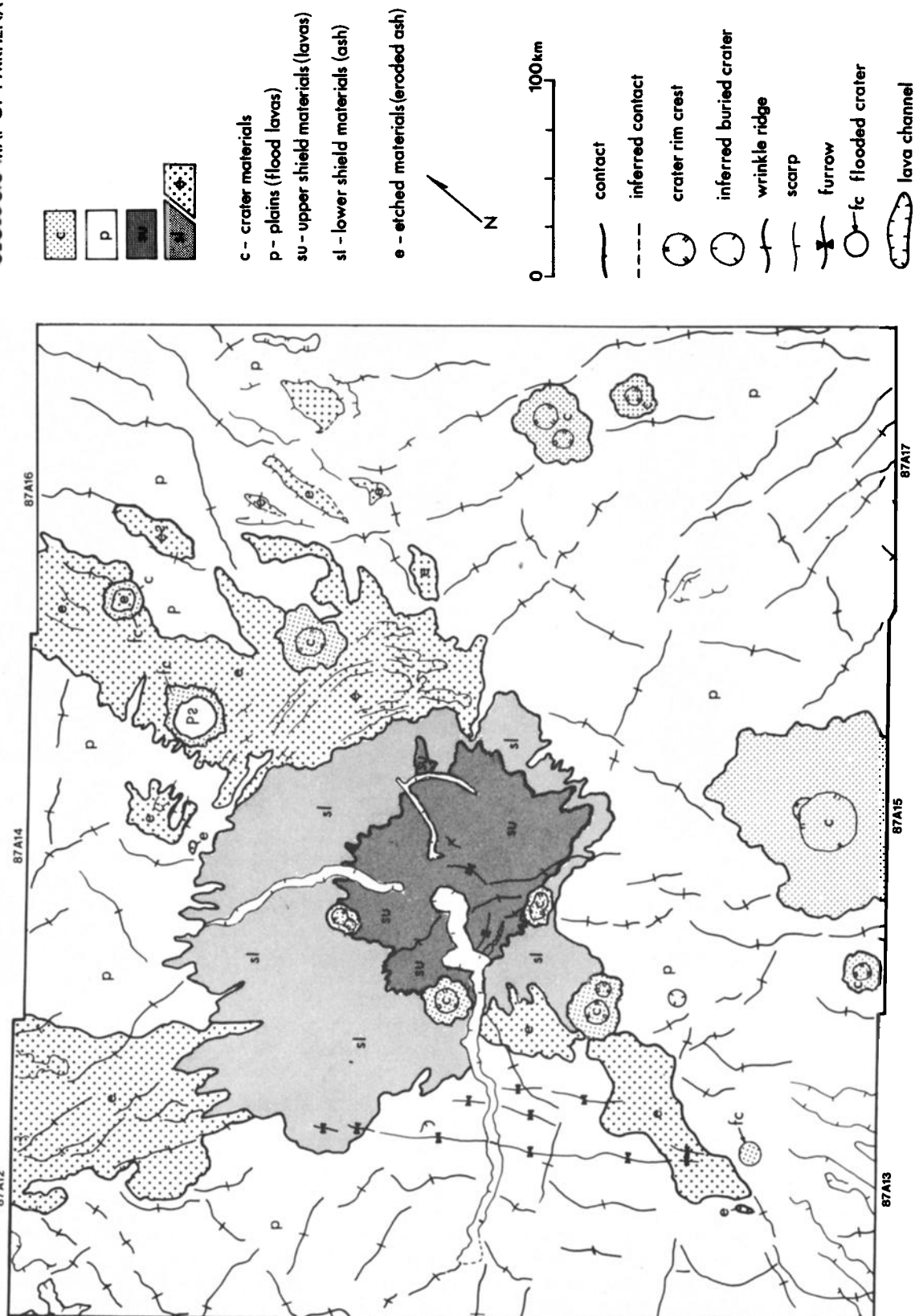


Fig. 8a. Geological sketch map of Tyrrhena Patera (corresponds to images shown in Figure 7).

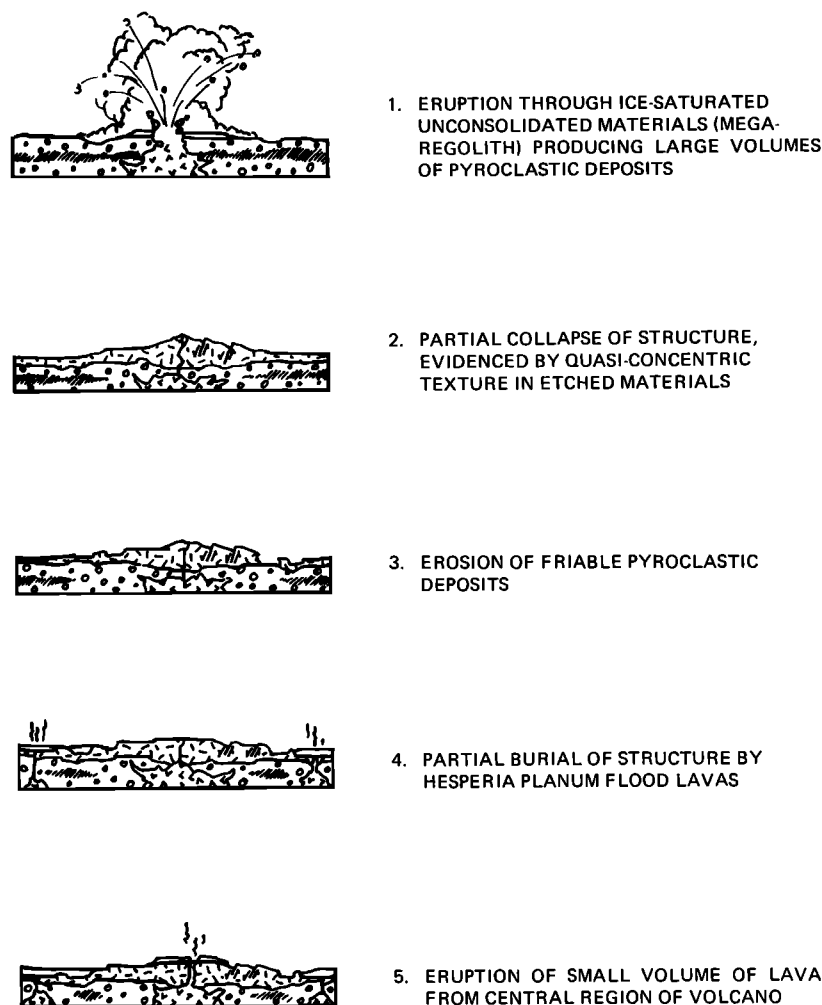


Fig. 8b. Sequential diagram of possible volcanic history for Tyrrhena Patera, based on the geological map of Figure 8a.

conservative estimate for the volume of the flows can be made.

Comparing these volumes to some typical rates of effusion of basaltic lavas on earth (Figure 14) gives some understanding of the enormity of the flows on Alba. Keeping in mind that there are many flows that appeared to be operating essentially concurrently, either extremely high rates of effusion must have been involved for the Alba flows, or the eruptions were of very long duration. From morphological considerations we favor the latter interpretation because on earth, high rates of effusion (i.e., flood eruptions) do not produce lava tubes, at least none that are preserved and recognizable. A long period of volcanic activity is supported by independent study of the tectonic history [Wise, 1976].

Thus we interpret the tube-fed flows on Alba to have developed through the eruption of basaltic lavas of moderate (approximately equal to Hawaiian) rates of effusion over many years of activity. Although eruption was nearly continuous, short periods of quiescence probably occurred, with renewed activity erupting lavas down the same tube system as in previous eruptions. From terrestrial analogy, some downward cutting of the tube system may have occurred; temporary blockages of closed tubes resulted in bifurcation of the system, all the while lengthening and extending the flow front.

Some of the tube-fed flows have domes or a series of domes

at the end of the flows. These may result from lava flowing within the tube system from the vent under hydrostatic loading. At the lower end of the flow the lava may rupture the roof of the tube as a pseudovent, resulting in local outpourings of flows to build up the domes. Similar features have been described for lava tube systems associated with volcanoes on earth [Greeley and Hyde, 1972]. Because most of the flow was contained within the tubes, heat loss was minimal; thus as long as eruption continued and a slope existed, the tube networks would continue to grow. Tube-channel flows occur west of the ring fracture and in a smaller area north of the ring fractures. These flows consist of a complex series of channel-fed and tube-fed anastomosing flows which represent multiple flow units.

Sheet flows constitute the most conspicuous type of lava flow on Alba Patera. Although most of these flows appear to originate near the ring fracture, the actual vents for the sheet flows are not visible and may have been buried by their own products. Typically, the sheet flows lack flow surface structures such as lava tubes, lava channels, and flow festoons but form multiple, overlapping lobes that have fairly level surfaces. Their closest analogs are the Imbrium flows on the moon and possibly the flood-type basalts of the Columbia River Plateau. Sheet flows are best developed northwest of the ring fracture, although they also occur within the ring frac-



Fig. 9a. Plains units in the Cydonia region (38°N, 12°W) showing small (approximately a few hundred meters) conelike features, most of which have summit craters; these and similar features elsewhere on Mars are interpreted by most investigators to be pyroclastic cones of volcanic origin (Viking Orbiter 1 image 72A02).

ture, where some appear to have originated from the central caldera complex. A particularly long (more than 270 km) flow occurs in the southwest part of the volcano. The relationship between rate of effusion and length of lava flows (non-tubefed) suggests that the sheet flows involved high rates of effusion, perhaps comparable to flood eruptions.

The remainder of the flows that make up Alba Patera are undifferentiated. Although many of these flows are fed by tubes and channels, they differ by having surfaces of rugged relief and flow margins that are indistinct. From superposition relations and their degraded appearance the undifferentiated flows at the distal ends of Alba Patera may be the oldest flows. The undifferentiated flows within the ring fractures are fresh appearing and have well-preserved channels with levees. Many of the flows originate from the central caldera complex, and some are superimposed on the rim of the caldera; other flows are cut by the caldera fracture, indicating an essentially contemporaneous eruption-caldera formation sequence. Except for the obvious flows that originate from the caldera, it is not possible to determine the source vents for most of the flows and flow units. For such an enormous structure it is rather surprising that subsidiary vents are not present, or at least more obvious. Either the subsidiary vents are buried by flows or they do not exist.

The styles of volcanism represented by the various flows suggest that Alba Patera involved both flood-type eruptions from a central region and Hawaiian styles over very long durations. Toward the end of its volcanic activity, Alba was subjected to tectonic deformation related to the presence of the Tharsis rise. Whether the cessation of volcanism with the tectonic activity is cause or effect cannot be determined. Given the enormous volumes of lava composing Alba Patera and the supposed very long period of volcanic activity, unanswered questions arise as to the nature of the magma chamber and the mechanism(s) involved in producing the magmas.

b. Highland patera. This category of central volcano, defined by *Plescia and Saunders* [1979], includes Tyrrhena Patera, Hadriaca Patera, the Amphitrites Patera complex, and possibly a feature in the northern hemisphere informally named Tempe Patera [*Plescia and Saunders*, 1979]. These features are all distinguished by their low profile, radial channels, and complex central calderas. They are relatively old, and all occur in or near the cratered terrain. Of the features that constitute this group, Tyrrhena Patera (Figure 7) has the best photographic coverage and can serve as the type example.

Figure 8 is a geological sketch map for Tyrrhena Patera. The oldest photogeological unit, termed etched materials, displays a rough, irregular topography strongly suggestive of erosion. This unit is embayed or cut by all other stratigraphic units. In one region north of the Tyrrhena structure a concentric structural grain is evident that is not expressed on the surrounding plains and which may represent the formation of an early caldera. The lower shield materials of Tyrrhena Patera directly overlie the etched materials. The lower shield material shows a markedly radial texture resulting from canyons that emanate from the center of the structure. On the basis of their intimate association with the volcano and apparent low erosional resistance, both the etched and lower shield materials are interpreted to be ash flows (discussed below) emplaced early in the shield building stage of Tyrrhena Patera. The upper shield materials show a smooth surface, uncut by radial channeling but containing wrinkle ridges similar to those seen on the surrounding plains. This unit is interpreted as lava flows that cap the previously erupted ash flow deposits. Although age relations are not clear, the surrounding Hesperia Planum lava plains were emplaced contemporaneously with the late stage lavas of Tyrrhena Patera. The sinuous channels originating near the summit caldera cut all the volcanic units and are interpreted to be lava channels that repre-

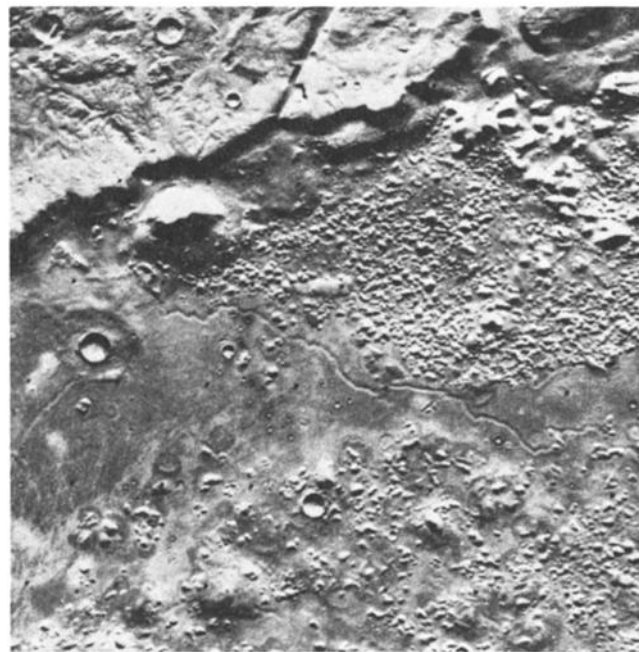


Fig. 9b. Plains in the Cydonia region (40°N, 9°W) showing a sinuous channel that may be a lava channel analogous to lunar sinuous rilles; the smooth material flanking the channel and mantling the adjacent terrain could be lava deposits associated with the channel, as observed in lava rivers on earth and interpreted on the moon (Viking Orbiter 1 image 72A11).

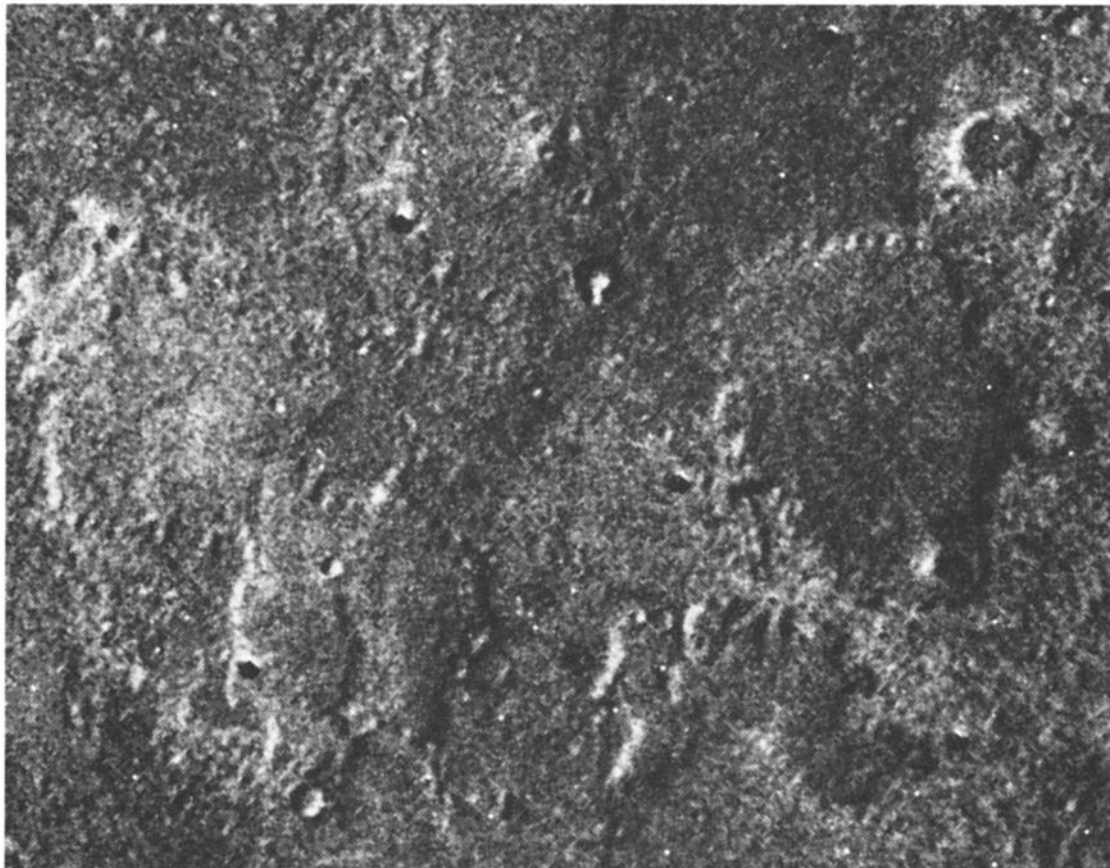


Fig. 10. Small features (a few kilometers across) in southern Chryse Planitia interpreted to be 'low shields' or small volcanic constructs. These and similar features are at the threshold of detection on high-resolution Viking orbiter images, and their interpretation as volcanic is open to question (Viking orbiter image 6A36).

sent the waning stages of Tyrrhena shield building. Some of these channels may have emplaced lavas in the surrounding plains.

An interpretation of the volcanic history of Tyrrhena Patera (Figure 8) can be derived from its morphology and the geological relationships shown on the map. The preeruption setting was heavily cratered terrain [Scott and Carr, 1978] on the outer rim of the Hellas Basin. Following the period of heavy impact cratering a Martian 'megaregolith' was probably generated. Integrated networks of fine scale channels of probable fluvial origin suggest the presence of surface water that sank into the regolith to produce a substantial groundwater zone [Carr, 1979]. During this stage in the evolution of Mars, heating, expansion, and mantle differentiation occurred [Toksoz and Hsui, 1978], leading to early stage volcanism. Eruption through the water-charged regolith would have resulted in extensive phreatomagmatic eruptions of ash, the volumes of which would be dependent upon the magnitude of the event(s). From lunar experience the concentric fractures generated by large impact basins are frequently the loci for volcanism. Thus it is reasonable to expect similar vents positioned around the Hellas Basin on Mars, as proposed by Peterson [1978a] and shown in Figure 1. However, unlike the anhydrous moon, early basin-associated volcanism on Mars probably involved phreatic eruptions.

The continued eruption of ash from essentially central vents built extensive ash sheets and 'ash shields' around the most active vents. The emplacement of the ash was followed by partial collapse, perhaps due to withdrawal of support result-

ing from eruption, and the formation of quasi-concentric fractures, analogous to caldera formation in terrestrial ash flow volcanism.

It has been proposed [Reimers and Komar, 1979] that some of the channels on Tyrrhena Patera are primary features formed by ash eruptions; however, the mechanics for the formation of such large features remain open to question. We prefer to interpret the radial texture of lower shield materials and 'etched' parts of the ash sheet to be the result of erosion by wind, water, and mass wasting. The large sinuous channel-like structures originating near the summit of the volcano and merging with the surrounding plains are interpreted to be lava channels that have, in part, supplied some of the plains lavas to Hesperia Planum. By this time, planetary heating had reached a high level, and volcanism was occurring through the eruption of extensive flood lavas in many parts of Mars (Figure 3). Flood lavas of Hesperia Planum encroached and partly buried the flanks of Tyrrhena Patera, as seen by contact relations where the flows enter the erosional channels and other depressions. The last stage in the volcanic history of Tyrrhena Patera is the eruption of a small volume of lava in the central region of the volcano, producing the upper shield materials, and lava channels continued to supply portions of the surrounding plains materials from the Tyrrhena summit region.

Although the relations of the other highland patera are not as clear as those of Tyrrhena Patera, from the available images they have nearly the same geomorphology and are therefore considered to represent similar types of volcanism.

TABLE 3. Classification of Named Martian Volcanic Features

Name	Location	Type	Area, 10 ⁴ km ²	Note
Alba Patera (1)	40°, 110°	Alba	113.0	1
Albor Tholus (2)	19°, 210°	dome	1.94	2
Amphitrites Patera (3)	−59°, 299°	highland patera	6.55	3
Apollinaris Patera (4)	−8°, 186°	shield	5.35	4
Arsia Mons (5)	−9°, 121°	shield	33.3	5
Ascraeus Mons (6)	11°, 105°	shield	14.0	5
Biblis Patera (7)	2°, 124°	shield	1.18	6
Ceraunius Tholus (8)	24°, 97°	dome	1.01	2
Elysium Mons (9)	25°, 214°	shield	13.1	7
Hadriaca Patera (10)	−30°, 267°	highland patera	9.24	3
Hecates Tholus (11)	32°, 210°	shield	2.59	6
Jovis Tholus (12)	18°, 118°	shield	0.29	6
Olympus Mons (13)	18°, 133°	shield	37.4	5
Pavonis Mons (14)	0°, 113°	shield	15.0	5
'Tempe' Patera (15)	44°, 63°	highland patera	1.54	3, 8
Tharsis Tholus (16)	13°, 91°	dome	1.68	2
Tyrrhena Patera (17)	−22°, 254°	highland patera	3.94	3
Ulysses Patera (18)	3°, 122°	shield	0.85	6
Uranius Patera (19)	26°, 93°	shield	2.71	6
Uranius Tholus (20)	26°, 97°	dome	0.38	2

Numbers in parentheses after name are keyed to Figure 1.

¹Unique volcanic structure on the terrestrial planets.

²May represent buried older shields or different style of volcanic activity.

³Interpreted as ash shields with less lava present.

⁴Oldest lava shield recognized on Mars; occurs on plains/uplands boundary.

⁵Main shields of the Tharsis complex.

⁶Shields that have been partly buried by surrounding plains lavas.

⁷Slightly steeper sides than the Tharsis shields.

⁸Informal name, see *Plescia and Saunders* [1979].

c. *Shield volcanoes.* The impressive shield volcanoes of the Tharsis region provided the first clear evidence from Mariner 9 data that volcanism was important in the evolution of Mars. This category of volcano is characterized by central vent(s) often in the form of complex calderas, and flank slopes of a few degrees. As is true of shield volcanoes on earth, the Martian shields are composed of hundreds of overlapping flows and flow units, many of which were fed through lava tubes and channels. The presence of lava tubes and channels is often cited as evidence for rock compositions in the general basaltic range, although in principle, any lava having similar rheological properties to basalt could produce similar morphologic features.

Martian shield volcanoes occur in three areas: the Tharsis region, the Elysium region (Figure 15), and a region southeast of Elysium (Apollinaris Patera). Even on Mariner 9 images the Tharsis shields were seen to be enormous by terrestrial standards, and the Viking pictures show them to cover even larger areas than was previously suspected (Figure 2). Many of the flows surrounding the shields are now seen to originate from the shields (Figure 16), giving, for example, Olympus Mons a basal diameter of 600 × 800 km. In other cases the flanks of some of the shields are buried by lava flows from younger volcanoes or from unidentified sources.

Arsia Mons displays relatively well preserved flow features and is covered by good quality images; it is considered to be of 'intermediate' age [*Plescia and Saunders*, 1979]. As such, it can be taken as a type example Martian shield volcano. Arsia Mons displays a multiple-slope profile [*Carr et al.*, 1977] consisting of a steep inner zone extending from the caldera outward to 400 km, a second zone having a gentler slope and extending an additional 400 km, and then a broad, relatively flat zone that merges with the surrounding volcanic plains. Earth-

based radar data show the inner zone to have a gradient of 0.01 to 0.03 with a sharp topographic break to the lower gradient (0.002 to 0.006) of zone 2 [*Saunders et al.*, 1978]. The upper, steeper zone is characterized by narrow flows with channels, the next zone is composed of relatively wider flows with channels, and then as the flows grade onto the plains, they spread still wider and the channels disappear. These flows and the relations between morphology and slope are consistent with basaltic volcanism and are observed in the formation of shield volcanoes on earth.

On the basis of analyses of the stratigraphic relations and relative dating from impact craters, *Crumpler and Aubele* [1978] proposed an evolutionary sequence for Arsia Mons as follows: (1) construction of the main shield (built upon previously erupted lava plains), (2) parasitic eruptions on the northeast and southwest flanks, (3) volcano-tectonic subsidence of the summit, producing concentric fractures and graben, and (4) continued volcanism along a fissure or rift zone bisecting the main shield, flooding the caldera floor and covering parts of the upper flanks.

Although the details of the flows and the sequence of evolution vary from shield to shield, all appear to have undergone a complex history, and all share the same general morphology. Some of the central volcanoes originally described as tholii (domes) and patera can be reassessed via the improved Viking images as partly buried or modified shield volcanoes (Figure 11). For example, Biblis Patera and Uranius Patera in the Tharsis region are probably older shield volcanoes that have been mostly buried by subsequent eruptions from adjacent volcanoes, leaving only the steeper, upper flanks of the shield exposed. Similarly, the flanks of Jovis Tholus have been buried, leaving only the domelike upper part exposed. Thus caution must be exercised when interpreting compositions of

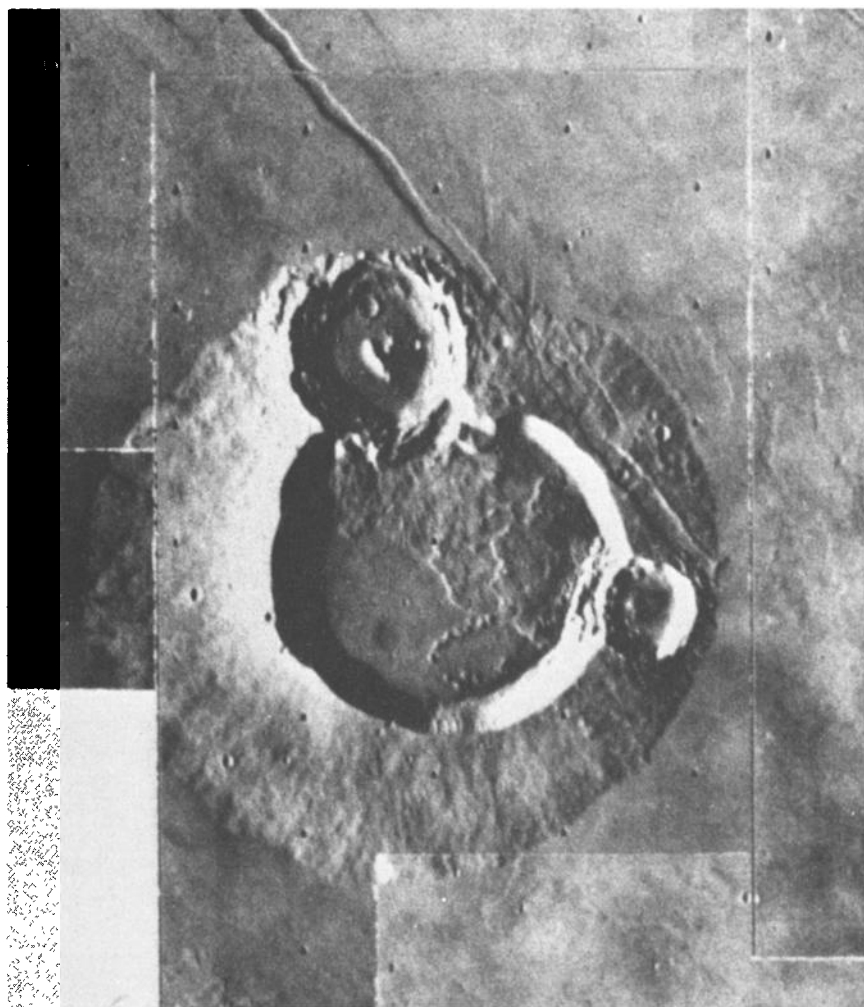


Fig. 11. Ulysses Patera, situated at 2°N , 122°W , is a nearly buried shield volcano; exposed shield volcanoes on earth and Mars often have an 'upper story' summit region surrounded by relatively steeper slopes than those that make up the lower parts of the volcano. Flooding of the volcano by younger lava flows, as seen here, buries the lower flanks but leaves the upper portion exposed. Some interpretations of volcanic style based on ratios of caldera diameter to volcano width were made prior to the recognition of these burial relationships and must be reevaluated. The occurrence of two large impact craters on the summit (but whose ejecta is also buried) attests to the great age of the volcano. Ulysses Patera is considered to represent early stage central volcanism in the Tharsis region. The circular outline of the caldera and the apparent lack of multiple flows on the floor suggest a simpler history than the histories of the younger Tharsis shields; note the small collapse pit and row of possible cinder cones on the floor. Area shown is about $170\text{ km} \times 190\text{ km}$ (Viking Orbiter 2 image 49B85).

Martian lavas on the basis of volcano profile. Steep slopes could result from one or more of the following factors: (1) increased proportion of pyroclastics in relation to lavas, (2) decreased rate of effusion, producing relatively shorter lava flows that would tend to pile up as steeper-profiled constructs, and (3) more silicic compositions that produce more viscous flows.

Variations other than flank profile among the shield volcanoes include the complexity of the central caldera, from relatively simple, such as Ulysses Patera, to complex, such as Olympus Mons (Figure 17), and the degree of tectonic modification, as with the complex ring graben and fractures of Pavonis Mons. Preservation of surface textures ranges from the relatively fresh flows of Olympus Mons to the degraded flanks of Apollinaris Patera. Despite these variations, all features classified as shield volcanoes are considered to represent a style of volcanism involving (1) a central vent system that produced multiple flow units, often fed through lava tubes and/or channels, (2) lavas that had rheological properties similar

to fluid varieties of basalt (i.e., pahoehoe), and (3) minimal pyroclastic activity, at least in the shield-building stage. Many of the Martian shields were subjected to tectonic deformation (principally summit collapse) in the waning stages of eruption, possibly related to emptying of the magma chamber. In addition, two of the shields, Olympus Mons and Apollinaris Patera, display basal scarps that cut the main structure. In the case of Olympus Mons, some later volcanic flows have buried parts of the scarp northeast and southwest of the volcano. The Olympus Mons scarp may be related to erosion of friable ash flows beneath the shield lavas [King and Riehle, 1974], erosion of a megaregolith substrate [Head *et al.*, 1976], initial eruption of lavas beneath an ice sheet, producing a large 'table mountain' [Hodges and Moore, 1979], or to general tectonic deformation associated with the Tharsis upwarp. The megaregolith erosion model of Head *et al.* [1976] may be particularly applicable to Apollinaris Patera, which is situated astride the northern-southern hemisphere scarp, a place where it is believed that ancient terrain has been gradually eroded away.

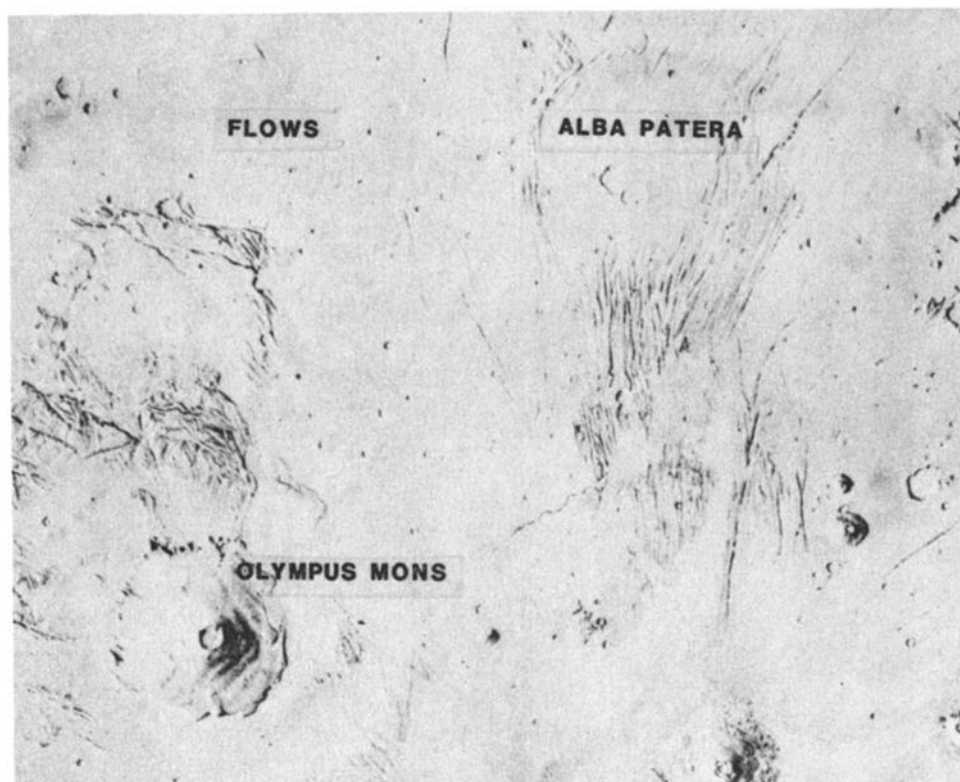


Fig. 12. Shaded airbrush relief map of Alba Patera (see Figure 1 for location). Sequences of flows can be traced more than 1500 km from the eruption center, making it the largest central volcano known on Mars, the moon, earth, or Mercury. Although some of the youngest flows are superposed over the ring fractures, most of the fractures cut the flows, indicating that most of the eruptive history occurred prior to tectonic deformation. (Base map from the U.S. Geological Survey.)

d. Dome volcanoes. Dome volcanoes, commonly named tholii on Mars, are characterized by steeper slopes (Figure 5) than are the shield volcanoes; in some cases, flanks exceed 8° in slope. The markedly greater slope of the domes in comparison to the shields is attributed to more viscous lavas, higher proportion of pyroclastics, lower rates of effusion, or some combination of these factors, reflecting a somewhat different style of eruption from that involved in the formation of the shields.

The surface texture of the domes ranges from relatively smooth and featureless, such as Tharsis Tholus, to high-relief lava flows (some with lava channels) as on Ceraunius Tholus. Nearly all of the features in this category have been partly buried by younger lava flows erupted from adjacent lava plains vents or mantled by younger deposits of nonvolcanic origin, suggesting an older geological age for the domes. Included in this category are various features that are possible composite cones, such as the feature identified in the Aeolis region [Greeley and Spudis, 1978].

e. Miscellaneous central vent volcanoes. Many areas on Mars outside the better known regions such as Tharsis display features suggestive of central volcanoes, including the Cydonia Mensae area [Hodges, 1979], Tempe region [Hodges, 1980], southwest Utopia Planitia [Woronow, 1980], and others. The features include fields of probable cinder cones near the margins of the Hellas and Isidis basins [West, 1974; Greeley and Spudis, 1978; Wood, 1979], table mountains [Hodges and Moore, 1979], and other structures resulting from phreatomagmatic eruptions [Frey et al., 1979; Allen, 1979]. As more high-resolution images become available for parts of Mars, more of the smaller features of probable and possible volcanic

origin are identified. Our mapping does not include all of these features, but some of the representative forms are shown in Figures 9 and 10 and briefly discussed. In addition, some parts of Mars appear to have been modified by volcanic and volcano-tectonic processes. For example, the floors of some impact craters may have been intruded by magma, producing uplifted floor-fractured forms [Schultz, 1978].

3. Volcanic Plains

Vast expanses of Mars are smooth on a kilometer scale, and plains of all types compose more than 60% of the surface area. It was recognized from Mariner 9 data that many of these smooth regions are of volcanic origin, although unequivocal photogeologic evidence for a volcanic origin (e.g., flow lobes) is present only in a few locations. This lack of evidence is partly a function of age and erosion but is mostly related to the style of plains emplacement.

Several criteria are used to classify plains on planetary surfaces as volcanic lava flows. Ideally, returned samples from geologically known areas will enable determination of the origin of plains units. In the absence of sample return, morphologic features permit the identification of some plains units as volcanic with varying degrees of confidence. These include lava flow fronts, embayment relations suggesting fluid emplacement, and mare-type 'wrinkle' ridges. All of these features are equivocal, however, and confidence in identification of these plains as volcanic decreases with decreasing occurrence. Although most controversial of the indicators, the presence of mare wrinkle ridges is frequently used to infer a volcanic origin for Martian plains (for example, Carr [1973] and others). There is still no general consensus as to how or why

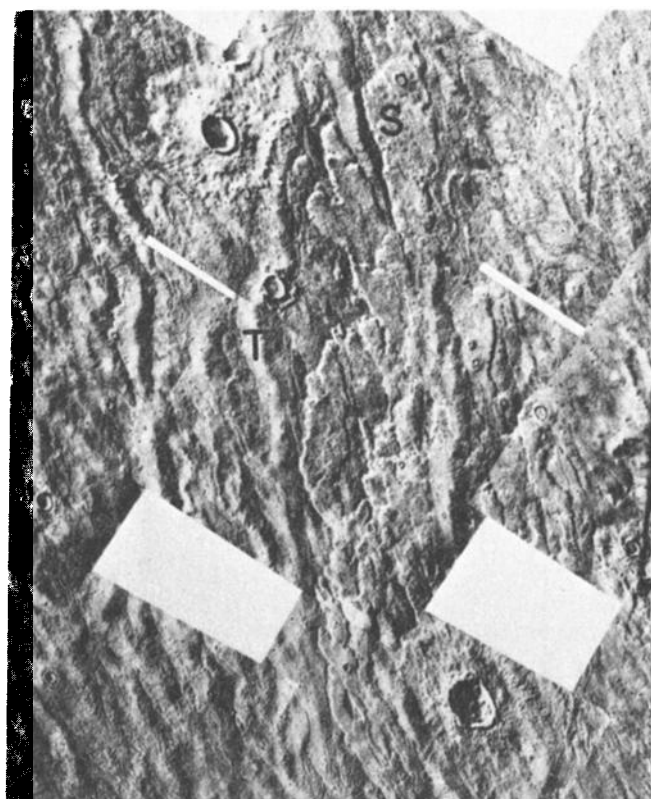


Fig. 13. Mosaic of the west flank of Alba Patera showing various tube-fed (T) and sheet (S) flows. North is toward the right (part of Viking orbiter mosaic 211-5065B).

the ridges form on the planets (also seen on the moon, Mercury, and possibly earth), and there are arguments to support both tectonic and volcanic hypotheses for some ridges as reviewed by Lucchitta [1976]. Mare-type wrinkle ridges display a broad arch upon which a crenulated upper portion is superposed. It has long been noted, however, that on the moon,

mare-type wrinkle ridges are confined exclusively to mare basalt units; where the ridges cross into the highlands, they have a simple scarplike surface expression. The surface morphology of ridges on plains is probably a result of the mechanical properties of the rock substrata: mare basalts are brittle, competent, well-lithified strata, whereas highland units are typically fragmental, brecciated debris. Thus the presence of mare-type ridges can be used to infer the presence of competent, brittle rocks. In the absence of sedimentary diagenesis this leads to a volcanic or impact melt origin for plains that contain ridges. An impact melt origin is unlikely for plains of regional extent and unrelated to large impact features, and so these plains are considered to be volcanic. In support of the use of wrinkle ridges as indicators of volcanic rocks, it is noted that the material filling the calderas of the large Martian shield volcanoes (which is almost certainly lava) has well-developed wrinkle ridges (Figure 17). It is conceivable (but highly unlikely) that some sedimentary diagenetic process could be responsible for generating regional deposits of flat lying sedimentary strata on Mars. Lucchitta and Klockenbrink [1979], however, note the absence of ridges in areas of presumed sedimentary layers such as in the canyon floors. Nonetheless, the conclusive identification of vast expanses of Martian plains as volcanic is open to question.

The volcanic plains of Mars have been categorized into four broad groups based on surface morphology: simple, complex, undifferentiated, and questionable flows. We emphasize that like volcanic plains classifications on earth, individual mapped regions on Mars may contain more than one type of unit; the category into which the region is placed is based on the prevailing unit it contains. Thus some 'simple' flow plains may have elements of complex flows present but only in relatively minor amounts. Each is discussed separately in the following section.

a. Simple flows. Simple flow plains display low relief, uniform regional extent, and numerous mare-type wrinkle ridges. The type area for simple flows is in Hesperia Planum,

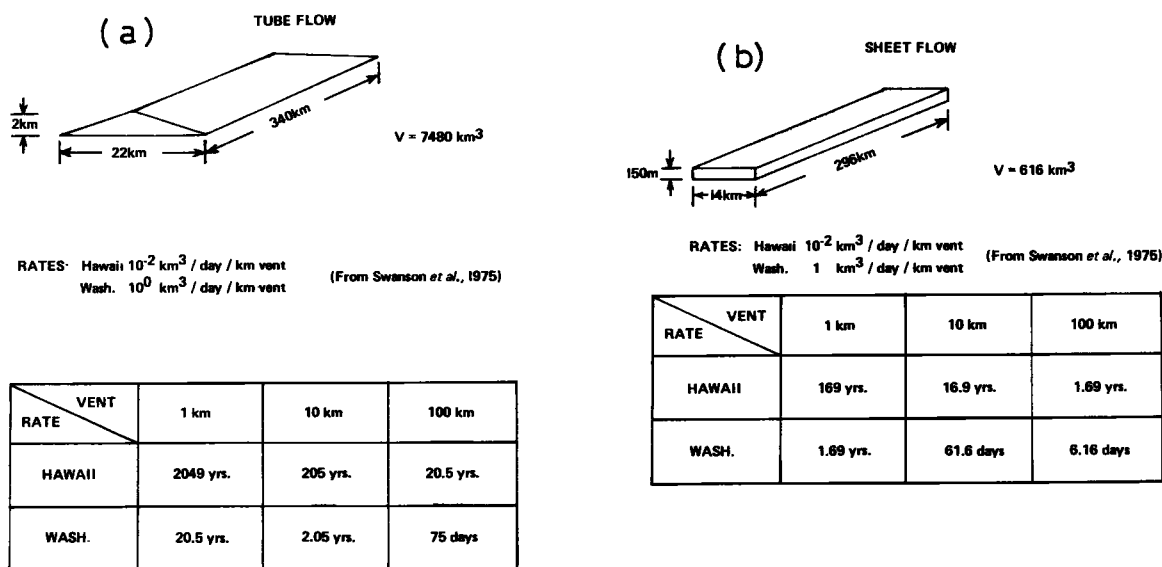


Fig. 14. Required eruption times at varying rates for the two primary types of Alba Patera lava flows. (a) The stylized geometry of a typical Alba Patera sheet flow as mapped on the southwest flank of the volcano. Shown is the duration of the sheet flow eruption based on a simple linear vent of 1-, 10-, and 100-km length for Hawaiian and Columbia River plateau eruption rates from Swanson et al. [1975]. (b) The stylized geometry of a tube lava flow on Alba Patera and typical emplacement times for a simple linear vent of 1, 10, and 100 km for Hawaiian and Columbia River plateau eruption rates from Swanson et al. [1975].

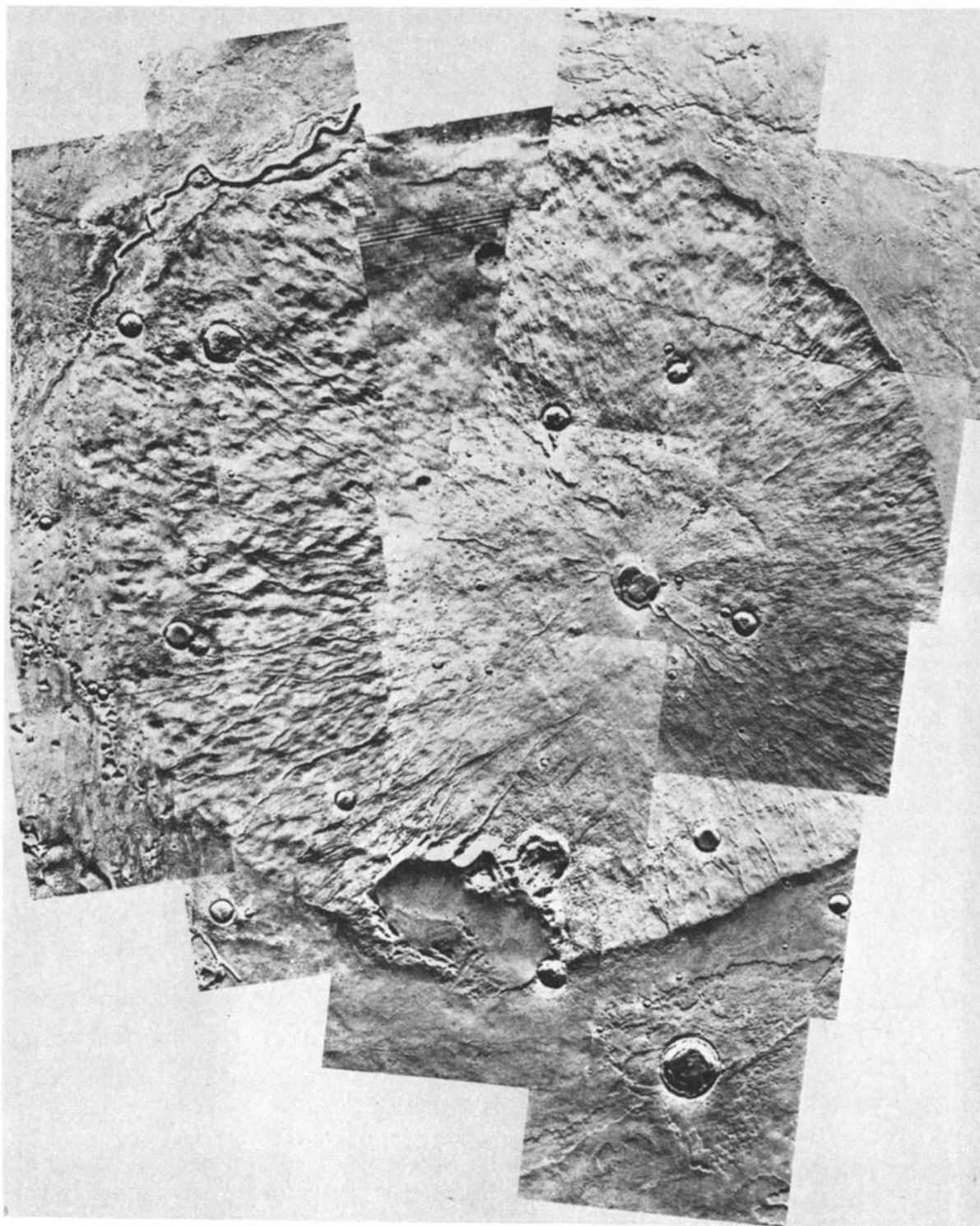


Fig. 15. Viking orbiter mosaic of Hecates Tholus, one of the prominent shield volcanoes in the Elysium region, more than 170 km across.

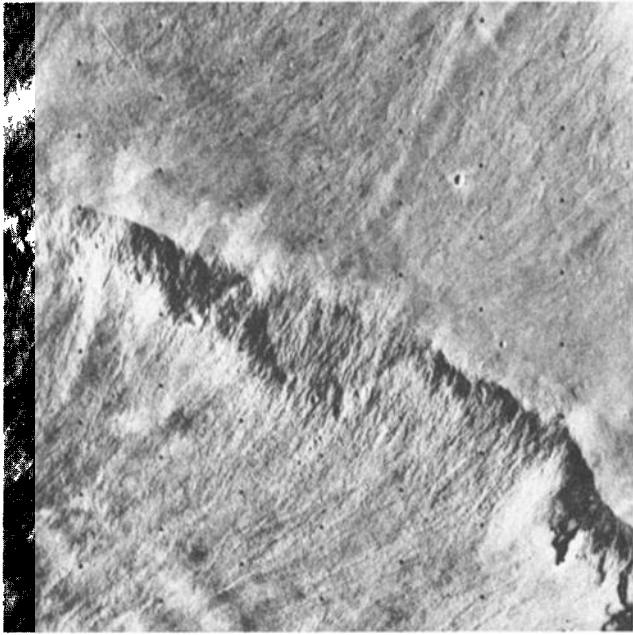


Fig. 16. Viking orbiter image of the northeast flank and basal scarp of Olympus Mons showing leveed lava channels and flows draped over the scarp and flowing onto the surrounding plains [from Carr *et al.*, 1977]. Area shown is about 80 km $\times$ 80 km (Viking Orbiter 2 image 47B25).

where the plains units appear to consist of a thick sequence of lavas. In some areas, circular wrinkle ridge structures suggest buried or flooded crater rims (Figure 18). Simple flows were probably emplaced as flood lavas, extruded at extremely high rates through long fissure vent systems that spread rapidly to resurface extensive regions. Such flows on earth typically form deep 'ponds' in low lying areas and lack apparent flow surface features such as lava channels and lava tubes but can show buried topography resulting from differential settling as the lava cools. On Mars, wrinkle ridges either may be contemporaneous with the flows, indicated by draping of buried crater rims, or may have developed in response to regional tectonic stress patterns at some time after lava emplacement, as seen in Lunae-Sinai Planum [Lucchitta and Klockenbrink, 1979]. We consider large portions of Martian plains that display only wrinkle ridges to be simple flows in the sense of Walker [1972] in that they represent high-volume, single-cooling units of great extent.

Simple flows appear to be primarily important as basin-filling and highland-resurfacing materials. All of the large Martian impact basins such as Hellas, Argyre, and Isidis are filled with simple flows. In addition, large, relatively flat lying regions of the southern cratered terrain hemisphere have been resurfaced by simple flows throughout Martian history. Thus although these flows are analogous in morphology and eruption style to lunar mare basalt flows, they also have been important agents in the volcanic resurfacing of the Martian highlands and appear to represent an areally significant fraction of the Martian surface.

b. Complex flows. Numerous plains units, particularly in the vicinity of the Tharsis upwarp, display a complex of overlapping flow lobes but typically lack wrinkle ridges (Figure 19). These include most of the youngest lavas on Mars, but the fact that small regions of flow-lobed lava plains can be identified also among the oldest plains on Mars [Greeley and Spudis,

1978] suggests that the presence of flow lobes is an indicator of mode of emplacement rather than a result of erosion and age. These plains have been called complex flows to distinguish them from the previously discussed simple flows, and they appear to result from a style of volcanism involving more protracted, sporadic, and lower rates of effusion. Although these flows are flood lavas in the sense of a relatively high (in comparison to Hawaiian, shield-building rates) eruption rate from a linear vent system, they are composed of compound, overlapping flow units of variable thickness and extent and are analogous to the compound flow (multiple-cooling units) units of Walker [1972].

Complex flows occur in numerous places on Mars but are much more limited in areal extent and in time than the simple flows. Small regions of older (lower Hesperian) complex flows may be identified around the margins of the Helles Basin, but they appear to play a minor role in the general basin-filling process. The major occurrence of complex flows is in association with the two major volcanic-tectonic complexes of Mars, the Tharsis and Elysium regions. In the Elysium region, complex flows make up most of the basal plains of the complex and typically embay the shields Hecates Tholus (Figure 15) and Elysium Mons. The Tharsis plains contain some of the most spectacular complex flows seen on the planet, with high densities of lava flow lobes of various lengths, heights, and widths. Extensive study of these flow scarps provides most of the rheological data currently available for Martian lavas [Schaber *et al.*, 1978].

Although complex flows have occurred throughout Martian history, most are of late (Amazonian) age. We interpret this as being indicative of a gradual change in the style of plains volcanism on Mars with time and consistent with a decreasing magma supply rate, since the cooling, thickening lithosphere prevents the release of large volumes of lava that would tend to produce simple flows upon reaching the surface, as was widely prevalent early in Martian history.

c. Undifferentiated flows. Plains on Mars that lack widespread features indicative of volcanism but which have a few, scattered features such as flow lobes are classified as undifferentiated flows. This category is loosely defined and may include plains related to aeolian or periglacial processes, particularly in northern latitudes. This classification includes numerous plains regions that are rather nondescript but are considered to be of probable volcanic origin on the basis of geologic setting and proximity to other well-defined volcanic regions. Therefore the inclusion of this category in estimates of volcanic areas should be made with caution, since it is difficult to assess the degree with which the assignment of a volcanic origin for some of these plains is justified.

The northern plains of Mars constitute most of this class and present a particular problem, but they are considered volcanic by most investigators [e.g., Guest *et al.*, 1977; Scott, 1979]. In general, these plains possess few of the criteria for volcanic plains. Either these plains never had the typical Martian lava plains morphology, or they have been extensively modified by processes peculiar to high northern latitudes such as periglacial processes [Carr and Schaber, 1977]. In several places, wrinkle ridges and occasional flow lobes are seen, as in the Ismenius Lacus region [Lucchitta, 1978]. Although this is not conclusive evidence for a volcanic origin for all the northern plains, we consider that the vast majority of these plains are at least underlain by volcanic substrata. This problem is in part linked with the general problem of the northern-southern

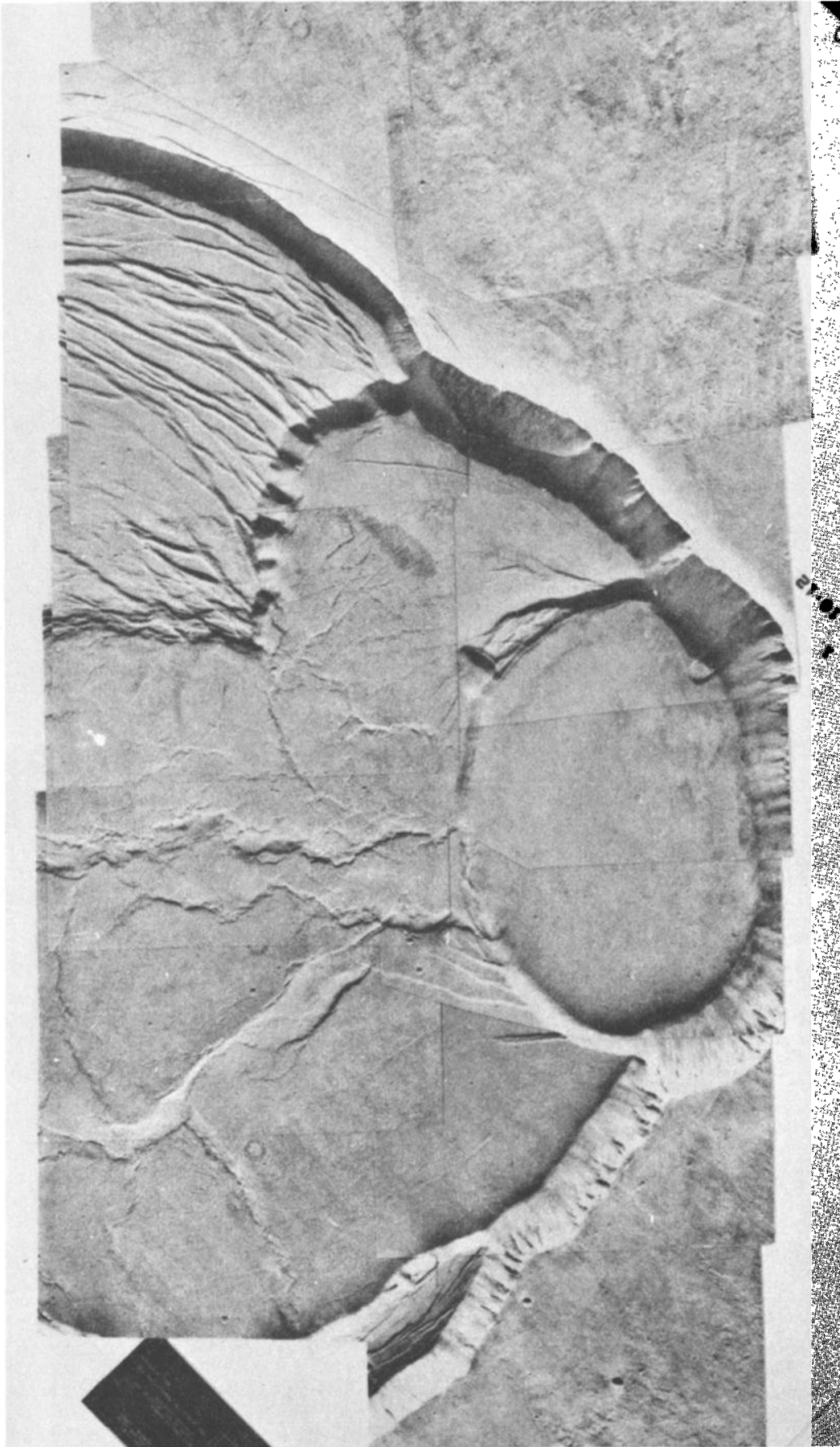


Fig. 17. Viking orbiter mosaic of part of the Olympus Mons caldera, showing extensive mare wrinkle ridges in one of the floor units. These ridges are typical in morphology to those that occur on lunar maria and on simple flow volcanic plains on Mars. The caldera floor materials here are almost certainly volcanic in origin and probably represent basalts that were 'ponded' as a relatively thick cooling unit. The occurrence of mare-type wrinkle ridges here in the caldera lends credence to the interpretation of mare ridges in plains as indicative of thick, ponded lava flows.

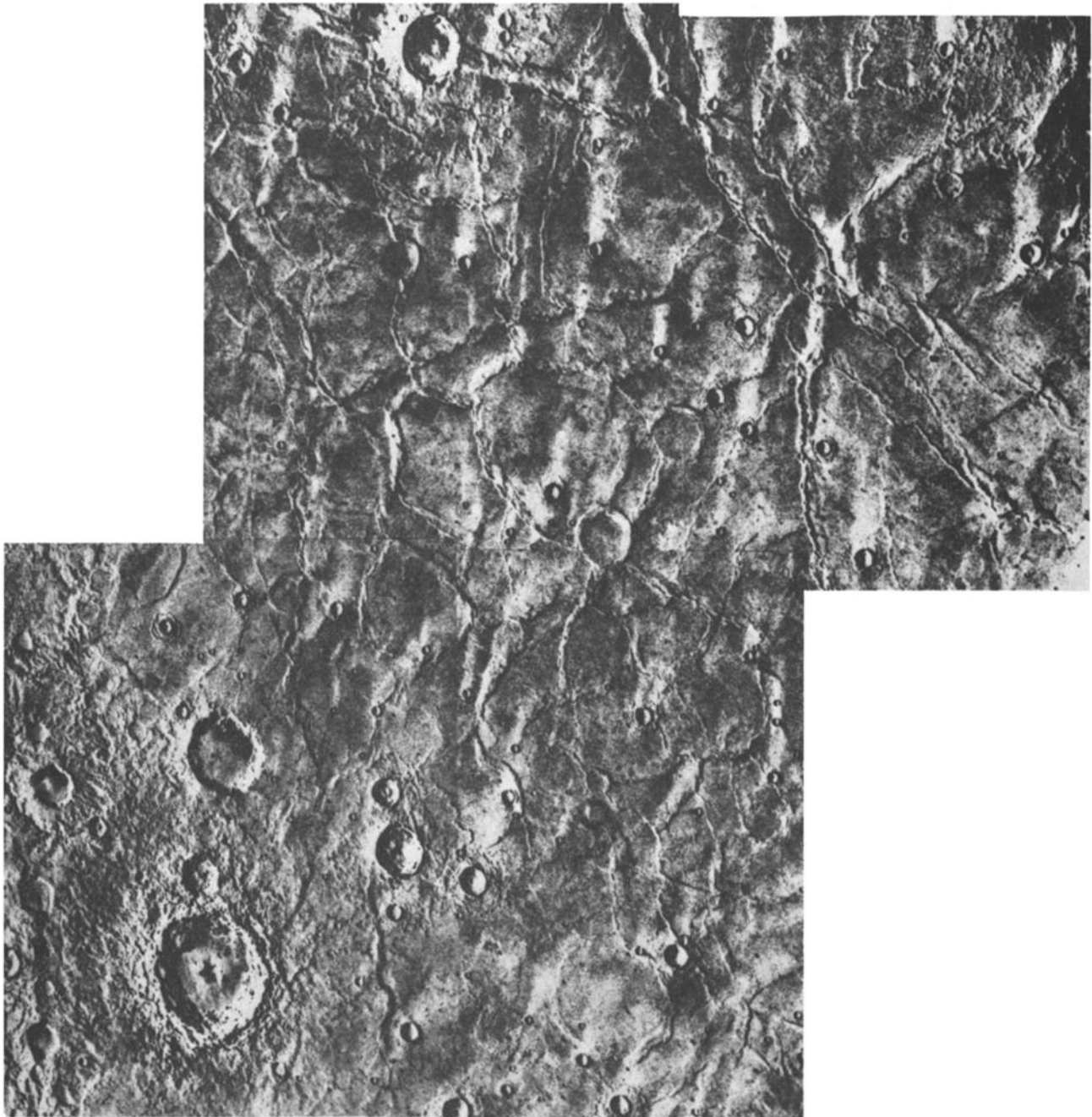


Fig. 18a. 'Simple lava flows,' in Hesperia Planum, showing typical surface with numerous mare-type ridges and possible buried craters; these flows are interpreted to have been erupted as flood lavas that produced thick, ponded units.

hemisphere dichotomy [e.g., *Malin et al.*, 1978], and the final resolution of that general problem is probably not immediately forthcoming.

Some of the unusual surface textures exhibited in the northern plains have been attributed to periglacial processes, or other processes that have modified the basic deposits. However, some of the textures described as modified may in fact represent primary lava flow textures. Figure 20 shows some of the Martian textures in question, compared to lava flow texture at the Amboy lava field. The terrestrial example consists of numerous basaltic flow lobes, tumuli, and depressions giving a local surface relief up to 10 m; low lying areas are filled with windblown sediments to produce a mottled appearance. Although different from Mars in scale, the general processes,

i.e., volcanic and aeolian, may be comparable. Thus we suggest that some of the textures observed in the northern plains, previously proposed as resulting from modification, may be primary lava flow surfaces.

Undifferentiated flows also occur around the margins of some of the major volcanic complexes such as Tharsis and Elysium (Figure 2). Although unequivocal evidence for a volcanic origin for these plains is lacking, there is no compelling evidence for an origin by other processes either. Because in many regions the plains grade into surrounding units confidently identified as volcanic, it is inferred that the vast majority of these plains are related to the general volcanic resurfacing associated with the construction of the major volcanic centers. The age relationships of these peripheral plains tend

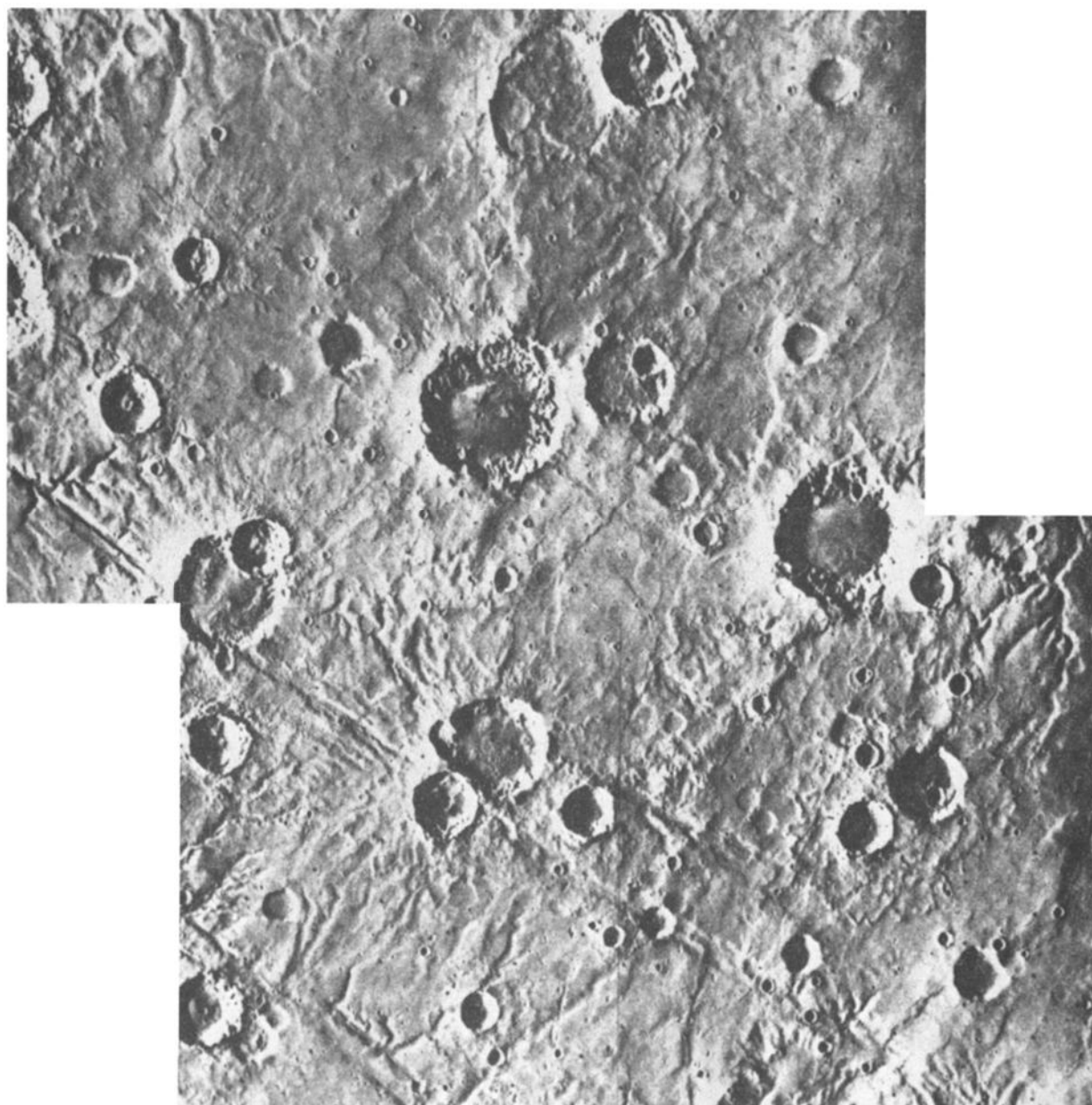


Fig. 18b. 'Plateau plains' units near the Argyre Basin.

to support this interpretation, since most appear to be synchronous with the confidently identified volcanic plains.

d. Questionable plains. Questionable plains include units extensively modified by erosion, fracturing, and mass wasting processes. Some of these units, such as heavily fractured plains peripheral to the Tharsis upwarp, are almost certainly older volcanic plains units that have been modified by faulting. Others such as the enigmatic aureole deposits of the Tharsis volcanoes [Hodges and Moore, 1979; Lopes *et al.*, 1980] are of more uncertain origin and could include volcanoclastic debris, such as mud and ash flows. The surface morphology of such rocks as seen from orbit is simply unknown, and many of the plains included in this category may not be volcanic. Conversely, this category may contain volcanic materials that are not lavas, such as the finely sculptured deposits southwest of Olympus Mons that may be ash deposits [Ward, 1979].

Numerous areas of questionable plains occur around the

Elysium complex and have been modified by complex tectonic and channeling processes. In analogy with the undifferentiated plains discussed above, these areas are considered to be volcanic on the basis of proximity to surrounding Elysium lava flows.

Because questionable units are typically so heavily modified as not even to permit the gathering of crater statistics, in many cases their relative ages are unknown. The uncertainties in age and the uncertainty of volcanic identification have precluded the inclusion of this unit on the volcanic history diagram (Figure 3).

In summary, plains constitute the majority of the surface units on Mars (Table 2). Even if plains in the undifferentiated and questionable categories are excluded from the sequence, volcanic plains are by far the predominant volcanic feature on Mars. Furthermore, many of the other plains on Mars may be deposits of volcanic ash. The possibility of phreatic activity on Mars, given the presence of water in the regolith and/or on

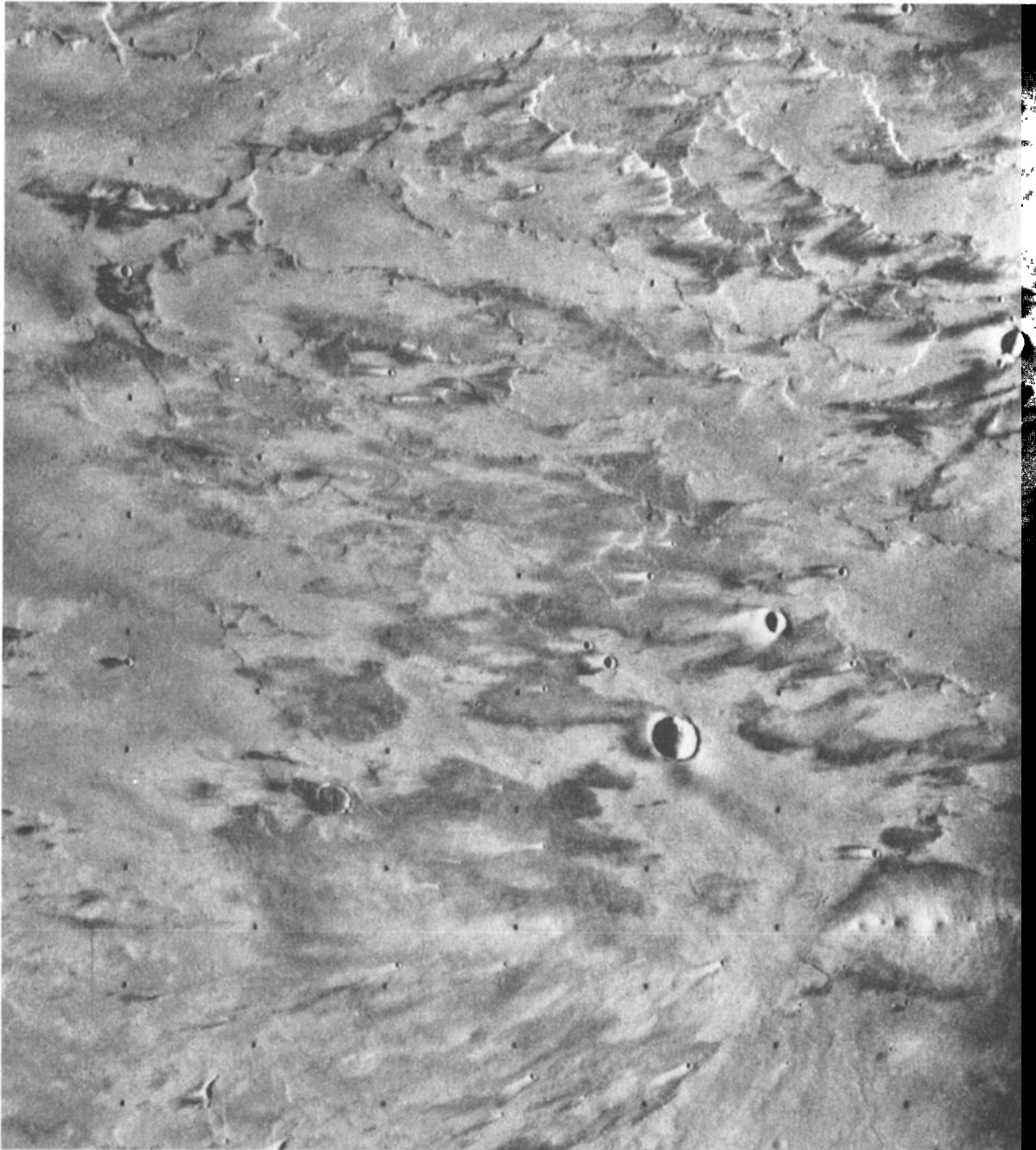


Fig. 19. Type area for 'complex flows' (Tharsis region; -6° , 131°W) displaying high density of overlapping lava flow lobes. Albedo markings are probably due to aeolian sediment. Area shown is 190×220 km (Viking Orbiter 2 frame 42B32).

the surface, has been previously discussed [e.g., *King and Reihle*, 1974], and Viking results have enhanced the possibility. Thus it is likely that extensive volumes of ash have been produced in the history of Mars. Unfortunately, photogeologic and other remote sensing criteria for the recognition of ash deposits have not been established but remain an area for future research.

E. VOLCANIC HISTORY

Using the classification scheme described above, volcanic features on Mars were mapped at a scale of 1:25,000,000 using Viking orbiter images. This mapping (Figure 2) is an update of earlier mapping [*Spudis and Greeley*, 1977; *Scott and*

Carr, 1978] based on Mariner 9 images and early Viking data. We point out that Viking image resolution and quality are not uniform over the planet and the identification and mapping of volcanic features is largely a function of the coverage. Medium-resolution (~ 50 m/pixel) images were used in most cases, supplemented by higher-resolution pictures for detail. Furthermore, the mapping consists mostly of large-scale features and does not include detailed mapping of all small features such as individual cinder cones and possible table mountains. The northern plains of Mars constitute the largest area not having good quality images, and consequently, the knowledge of their history is very poor.

Relative dating of the mapped volcanic features and terrain

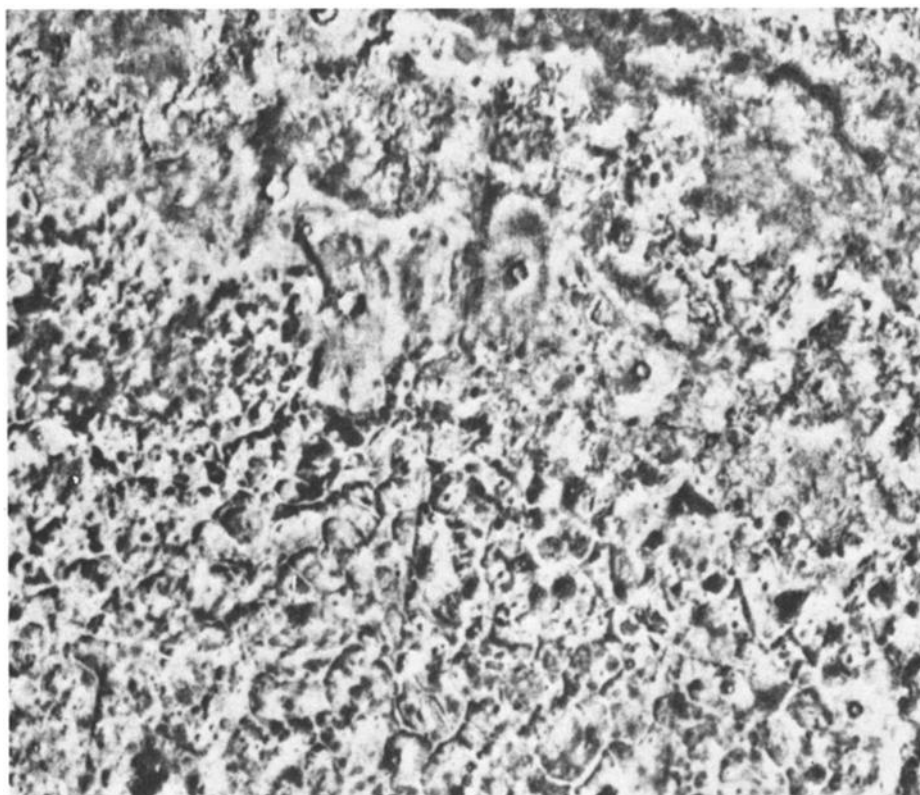


Fig. 20a. Portion of the Martian northern plains near Viking Lander 2 site showing hummocky surface that may represent a primary volcanic surface similar to that shown in Figure 20b (area shown is about 100 km across; Viking orbiter mosaic 211-5072, part).

is based on superposition, crosscutting relations, and the number of superposed impact craters. These are keyed to the formal time-stratigraphic sequence derived by *Scott and Carr* [1978] for Mars: the Amazonian system (youngest), Hesperian system (intermediate age), and Noachian system (oldest). Crater frequency distributions are derived primarily from *Condit* [1978] and are based on craters in the 4- to 10-km-size range. The rationale is that craters of this size are large enough to be visible despite weathering and erosion but still small enough to allow sufficiently large numbers to be statistically significant over large areas. Crater counts for the central vent volcanoes are from *Plescia and Saunders* [1979]; however, caution must be exercised in interpreting the results, since the areas 'counted' for each volcano are very limited.

The volcanic history of Mars is summarized in Figure 3. In this diagram the relative amounts of the planet covered by volcanic materials during a given stratigraphic interval in the geologic past are represented by envelopes that have areas proportional to the currently exposed areas on the planet as a function of relative time. Therefore these estimates should be considered minimum values in that portions of the older units may be buried by younger units and/or destroyed by erosional processes. It is evident that volcanism of one type or another has been active throughout the history of Mars.

The oldest units that are identified as volcanic on Mars are the plateau plains (S_1 [Wilhelms, 1974; Greeley and Spudis, 1978]). These plains take the form primarily of simple lava flows, with flow fronts rarely being present. The bulk of these flows were extruded more or less contemporaneously with the final stages of heavier bombardment, but most appear to post-date the major impact basins Hellas, Argyre, and Isidis. It

should be noted that the original extent of these plains was probably much greater than is currently exposed, in part owing to burial by younger flows in the southern hemisphere but also to the obliteration of the cratered terrain by a process still obscure in detail. These episodes of extensive volcanism overlapped in time with extensive fluvial activity, as evidenced by the abundant small channel networks present in the Martian cratered terrain [Pieri, 1976]. The overlap of volcanic and fluvial activity would strongly favor the production of paeatmagmatic activity. Thus some of the extensive older undifferentiated and questionable plains may be ash sheets. Highland patera (hp), interpreted here as partly ash shields, are also contemporaneous with the plateau plains. As noted previously, the highland patera are concentrated primarily near the margins of the Hellas Basin, possibly related to ring fractures, and all are located in the cratered southern uplands hemisphere.

After most of the plateau plains activity had subsided, massive flood volcanism (S_2 , C_1) was active over much of the planet, resurfacing areas in Lunae Planum, Hesperia Planum, and large parts of the northern hemisphere (u). During this epoch the northern-southern hemisphere erosional scarp began forming, a process that probably continued well into the Hesperian age. Volcanism leading to the formation of the Alba Patera structure (AP) also became active, possibly owing to structural fracturing in the lithosphere antipodal to the Hellas Basin [Peterson, 1978b]. This volcanic center was to remain an active central vent volcano during most of Hesperian time. Basin-filling by simple flood lavas also began in this epoch, as evidenced by extensive flows in the Syrtis Major Basin. Although these are the only lower Hesperian basin-fill

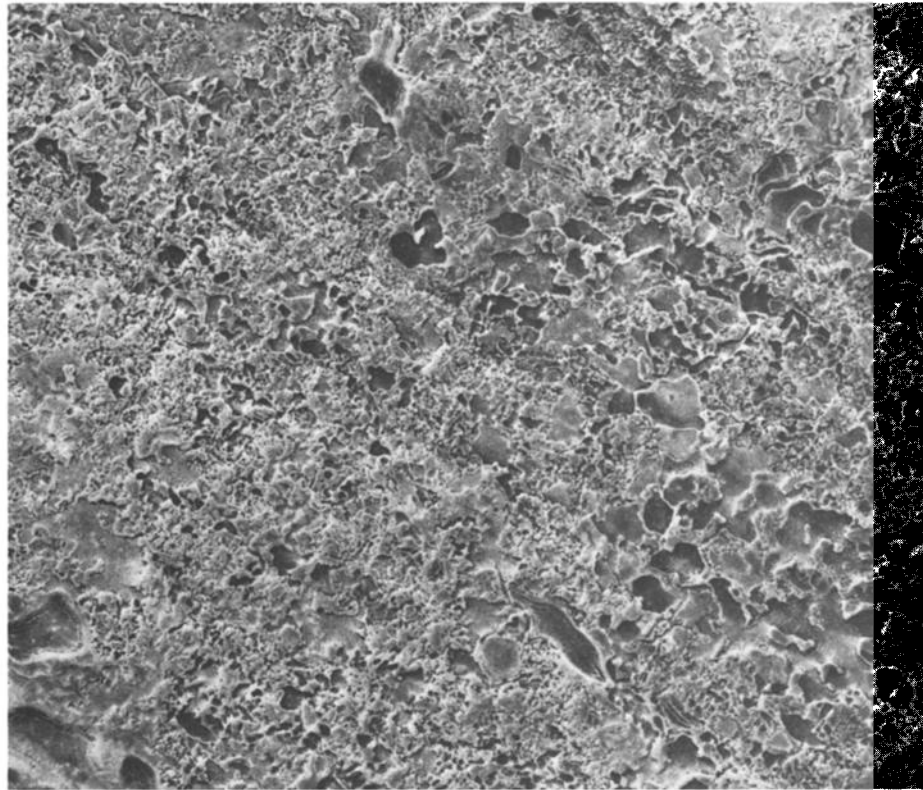


Fig. 20b. Hummocky basalt surface at Amboy, California (Mojave Desert); flow surfaces of this type result from numerous, coalescing lava 'toes' and 'tongues' to produce local relief up to 10 m; although about an order of magnitude smaller in scale, this primary flow surface may be similar to surface textures seen in parts of the northern plains of Mars. (U.S. Department of Agriculture aerial photograph AXL-1K-75.)

lavas currently exposed, it is likely that all the major basins were undergoing subsidence and volcanic filling at this time.

Most basin-filling activity was essentially completed by late Hesperian time. These plains (S_3) covered the interiors of the Argyre, Hellas, Isidis, and Chryse basins. This was a period when most of the northern plains had been emplaced and the rate of resurfacing by volcanism had become greatly reduced. It is also evident that central vent volcanism was becoming more predominant with the formation first of the Elysium volcanic centers and the early proto-Tharsis upwarp. The Elysium center consists of central volcanoes and a series of simple and complex lava flows, structurally uplifted to produce radial fracturing of a, by then, rigid lithosphere. This basic structure also applies to the proto-Tharsis volcanic center [Wise *et al.*, 1979b], although these units are now almost totally buried by younger volcanic flows.

The Amazonis system is primarily associated with the Tharsis volcanic center. Both central (sh, d) and plains (C_2) volcanism was important in constructing this vast volcanic region composed of shield volcanoes, domes, and complex flows. It appears that the bulk of volcanic activity was essentially completed in early Amazonis time and the extremely young flows (as determined by low crater densities [e.g., Schaber *et al.*, 1978]) constitute a very small fraction of the Martian surface.

The volcanic history of Mars as outlined above is consistent with a moonlike thermal history involving high rates of volcanic activity early in Martian history, gradually decreasing in volume of extrusion with a presumably cooling, thickening lithosphere as the planet evolved.

F. SUMMARY AND CONCLUSIONS

The volcanic history of Mars spans nearly the entire 'visible' history of the planet from the very earliest episodes of heavy impact cratering through to some of the youngest features, excluding the polar region and the currently active aeolian features. The styles of volcanism represented and the volcanic processes involved in the formation of the volcanic features are much more complex than those interpreted for the moon and Mercury and may equal those of earth.

Most of the volcanic activity on Mars appears to have involved eruption of mafic to ultramafic lavas—most probably basaltic. Evidence for this composition is derived from Viking lander results, remote sensing, and interpretations of landforms. Unlike the moon, where linear vents and flood-style volcanism predominates, Martian volcanism involves both linear vents and central vents that lead to the formation of extensive lava plains and a variety of central volcanoes ranging from extremely low profile form through classic shield volcanoes and steep-sided domes.

Evidence for phreatomagmatic activity includes both photogeology and compositional considerations. Many of the central volcanoes appear to be underlain by easily eroded material interpreted by many investigators to be ash. Proposed sequences of formation often begin with eruptions of magma through water-saturated megaregolith materials to produce ash deposits. With time, presumably the megaregolith is depleted in water near vent eruptive sites, and the volcanic activity transforms to effusion of lavas to build shield or dome volcanoes. The interpretation of dust in the atmosphere and the fines at the Viking landing sites as being composed partly of

palagonite lends credence to the possibility of extensive Martian phreatomagmatic activity.

Despite the advances in knowledge and the growth in the data base as a consequence of the Viking mission, there remain a great many questions regarding Martian volcanism:

1. What is the range of compositions of Martian volcanic rocks? Although substantial evidence points toward basaltic volcanism, do intermediate and silicic volcanoes also occur?

2. What is the range of absolute ages of Martian volcanoes? Although earliest volcanism occurs contemporaneously with the late stage heavy bombardment, typically accepted as ~4.0 aeons, what is the youngest volcanism? Is volcanism currently active?

3. Are ash flows (basaltic, silicic, etc.) present on Mars? Despite indirect evidence, photogeological criteria for the recognition of ash flow of any composition have not been defined for earth, let alone Mars; yet the identification of such flows would have an extremely important impact on the volcanic history for the planet.

4. Regarding the 'wrinkle ridge' problem, as discussed in section D3, the presence of lunar mare-type wrinkle ridges is commonly used as an identifying criterion for lava flows on the terrestrial planets. Yet, the evidence for this is indirect. Thus until the mechanism(s) for the formation of ridges is fully understood, the use of ridges to identify volcanic units is open to question.

5. What are the geochemical, petrologic, and geophysical properties of the magma sources on Mars? Nearly all questions regarding the subsurface and geophysical characteristics on Mars remain open.

The solutions to these questions are difficult and in many cases will not be forthcoming for many years. Questions 1 and 2 can be answered best with returned samples from carefully selected sites on Mars, although questions regarding composition might be answered to some extent with high-resolution remote sensing data from an advanced orbiting spacecraft. Questions 3 and 4 have the potential to be answered through a combination of photogeology, earth analog studies, and possibly laboratory simulations. Question 5 can probably be addressed only through the emplacement of instruments on Mars and experimental petrologic study of returned Martian lava samples.

The classification and mapping presented here should not be considered definitive. Just as the earlier studies of Martian volcanism based on Mariner 9 data were substantially revised by incoming Viking data, so too must this study be revised when the Viking data set is fully utilized. As noted above, this study was based on moderate-resolution Viking orbiter images and includes relatively few of the high-resolution images. We anticipate that detailed analyses of the high-resolution coverage will add substantially to the knowledge of localized volcanism (through the identification of cinder cone fields, etc.), which, when viewed globally, will lead to a much better understanding of the general volcanic history of Mars.

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